

# Quantum Mechanics as Multiphase 3D Continuum Mechanics

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*The collaboration is documented in Section 13.*

## Abstract

We formulate quantum mechanics as multiphase 3D continuum mechanics: an  $N$ -electron wave function  $\Psi = \sum_{i=1}^N \psi_i$  with unit-charge components  $\psi_i$  supported on non-overlapping domains  $\Omega_i \subset \mathbb{R}^3$  separated by Bernoulli free boundaries. Ground states minimise the standard Coulomb energy over both wave functions and domain decomposition, via coupled imaginary-time propagation, Poisson updates of per-electron Hartree potentials, and level-set evolution of the free boundaries. We refer to this real-space formulation as **RealQM**.

The framework is organised as a six-level hierarchy of reductions: parameter-free atoms (**L1**, total energies Li–Rn within  $\sim 1\%$  from minimum-energy electron packing); spherically-reduced inner shells (**L2**); pseudo-kernel atoms with effective charge and softening (**L3**); molecular and crystalline assemblies (**L4**); per-species reduced-model records (**L5**); and Brownian dynamics of  $\sim 10^6$  proteins on a single GPU (**L6**). Validation across chemistry:  $\text{H}_2$  at parameter-free L1 to 0.3% of Kolos–Wolniewicz total energy and 3% of experimental dissociation; closed-shell hydrides ( $\text{NaH}$ ,  $\text{BeH}_2$ ,  $\text{H}_2\text{O}$ ,  $\text{CH}_4$ ,  $\text{SiH}_4$ ,  $\text{GeH}_4$ ) within 3–9% across groups 1, 2, 14, 16; reactive chemistry (proton transfer, exchange); excited states (orthohelium); ionization energies. At L4 the framework supports **interactive protein folding**: polyglycine hairpins,  $\alpha$ -helices, and mini-proteins fold under a universal  $i \rightarrow i+4$  H-bond bias plus a single hydrophobic-pressure parameter. A four-system condensed-phase calibration ( $\text{NaCl}$  melt,  $\text{CO}_2$  dry ice, solid  $\text{N}_2$ , bare-nucleus He) captures non-dispersive cohesion and cleanly fails to bind at the dispersion-only endpoint. Applied to **liquid water**, the parameter-free model isolates the molecular driver of the two-structure (LDL/HDL) debate — the isotropic oxygen acceptor that permits hydrogen bonding is the same feature that forbids spontaneous tetrahedral (LDL) order — and places water in the correct small-viscosity, small-compressibility regime, the “regime not precise values” relevant to its macroscale incompressible, high-Reynolds behaviour. A set of **drug-mechanism reactive scans** (omeprazole,  $\beta$ -lactam, kinase warhead, GAD65) recovers the correct activation physics for covalent-addition mechanisms parameter-free.

Stability of matter for  $N$  non-overlapping atoms with kernel charge  $Z$  follows from the Hardy inequality applied to the global  $\Psi$  in a one-paragraph derivation, giving  $E \geq -CNZ^3$  — strikingly more compact than the standard Dyson–Lenard–Lieb–Thirring proof. The same multiphase Coulomb formulation extends to **atomic nuclei** in the pre-1932 proton–electron picture (neutron = proton + electron) with equal mass  $m_e = m_p = 1$ , through a three-level nuclear hierarchy: 3D polyhedral configurations for light nuclei (He-4 through C-12 within 25% of experiment); a single-shell packing model with proton thickness  $d = c/Z$  that reproduces the rising  $B/A$  curve to the Fe–Ni peak with two parameters; and an extension with electron-core expansion  $R_e^{\text{eff}}(Z) = R_e + \gamma \max(0, Z - Z_{\text{break}})$  that recovers the full experimental binding curve from H-2 to U-238 — all from pure Coulomb interactions, no strong-force postulate. The model predicts an electron-to-proton size ratio scaling as  $Z^{2/3}$ , the same exponent as the Weizsäcker surface term but emerging from inside-out geometric topology rather than from short-range force saturation.

The framework is implemented in  $\sim 8000$  lines of JavaScript and WebGPU compute shaders running interactively in a browser on a consumer laptop GPU — three orders of magnitude smaller and  $\sim 10^2$ – $10^4\times$  faster than mainstream quantum-chemistry packages. The implementation, validation suite, and curated public Gallery were developed through extended collaboration with an AI code assistant (Claude, Anthropic). We present the collaboration as part of the contribution: a worked case of how a single mathematical mind paired with an AI engineering partner can produce research-grade interactive scientific software at a scale that previously required a programming team.

**Keywords:** quantum chemistry; multiphase wave functions; free-boundary problems; Bernoulli condition; real-space methods; stability of matter; protein folding; nuclear binding energy; Weizsäcker formula; WebGPU; AI-assisted scientific software; human–AI collaboration.

## 1 Introduction

The  $N$ -electron Schrödinger equation has been the foundation of quantum chemistry for nearly a century. In its standard formulation, matter is described as eigenfunctions of an  $N$ -body Hamiltonian acting on antisymmetric wavefunctions in a  $3N$ -dimensional configuration space. The exponential scaling of this representation has been mitigated by approximate methods — Hartree–Fock, density functional theory, configuration interaction, coupled cluster — each making specific compromises between accuracy and computational cost [2, 3, 4]. Production-quality quantum chemistry packages comprise hundreds of thousands to millions of lines of code, draw on decades of accumulated machinery for basis sets, integral evaluation, and analytic gradients, and require dedicated computational clusters for systems of biological size.

**Two foundational positions.** Two quotations from the founders frame the conceptual position of this paper. The first is from **Dirac, 1929**, on the relation between physics and chemistry:

The underlying physical laws necessary for the mathematical theory of [...] the whole of chemistry are thus completely known, [and] the difficulty is only that the exact application of these laws leads to equations much too complicated to be soluble.

The second is from **Schrödinger, in a letter to Lorentz dated 6 June 1926**, sketching how he originally intended to interpret the wave function for  $N > 1$  electrons:

If we now have to deal with  $N$  particles, then  $\Psi(x_1, \dots, x_N)$  is a function of  $N$  variables  $x_1, \dots, x_N$  over  $N$  3D spaces  $R_1, \dots, R_N$ . Now first let  $R_1$  be identified with the real space  $\mathbb{R}^3$  and integrate over  $R_2, \dots, R_N$ . Second, identify  $R_2$  with the real space and integrate over  $R_1, R_3, \dots, R_N$  and so on. The  $N$  individual results are to be added after they have been multiplied by certain constants which characterise the particles. *I consider the result to be the electric charge density in real space.*

A third foundational position deserves to be set alongside the first two: that of **Walter Kohn** in his 1998 Nobel Lecture on Density Functional Theory [18], where the founder of the most successful working tool of standard quantum chemistry observed that

in general the many-electron wave function  $\Psi(\mathbf{x}_1, \dots, \mathbf{x}_N)$  for a system of  $N$  electrons is not a legitimate scientific concept when  $N \geq N_0$ , where  $N_0 \approx 10^2$ – $10^3$ .

Kohn’s statement is the working-physicist counterpart to Dirac’s: the configuration-space wave function loses its status as a scientific object already at small molecular scales. DFT’s response was a dimensional reduction to a single 3D density  $\rho(\mathbf{x})$  with an approximate exchange–correlation functional; RealQM’s response, developed below, is a foundational reformulation that never invokes the  $3N$ -dimensional object in the first place.

The three positions are not complementary; they are in tension. Dirac states that chemistry reduces to known physical laws; the only obstacle is computational — but he treats the  $3N$ -dimensional configuration-space form of the equations as fixed and concludes that the only relief is approximation. Schrödinger sketches the real-space alternative in which the  $N$ -electron wave function projects to an actual charge density in physical  $\mathbb{R}^3$ , recovering the direct physical meaning that  $\psi^2$  has for the hydrogen atom. Kohn, from the working physicist’s side seventy years later, observes that the  $3N$ -dimensional object Dirac took as fixed has, in practice, ceased to be a scientific concept already at small molecular scales — which is the diagnostic that Schrödinger’s letter to Lorentz had pointed toward all along. **The real-space-density interpretation was abandoned during the late 1920s under the dominance of the Bohr–Born–Heisenberg–Dirac Copenhagen interpretation**, in which  $\Psi$  on  $\mathbb{R}^{3N}$  became a probability amplitude rather than a physical charge density. The resulting Standard QM (StdQM) inherits the computational obstacle Dirac identified — solving an equation on  $3N$ -dimensional configuration space — and the chemistry-specific approximations introduced to make it tractable have, ever since, made chemistry look more autonomous from physics than Dirac’s position would suggest. RealQM, developed below, takes the choice made in 1926–1929 to have been the wrong choice: a real-3D-space formulation in the spirit of Schrödinger’s letter exists, dissolves the computational obstacle Dirac identified, and recovers the scientific status that Kohn observed the  $3N$ -dimensional wave function had lost.

The formulation developed in this paper, which we call **RealQM**, returns to the originally proposed real-3D-space picture. The  $N$ -electron wave function  $\Psi(\mathbf{x}) = \sum_{i=1}^N \psi_i(\mathbf{x})$  lives on physical  $\mathbb{R}^3$  as a sum of **one-electron wave functions**  $\psi_i$  on a non-overlapping decomposition of  $\mathbb{R}^3$  into domains  $\Omega_i$ , with each  $\psi_i^2$  a unit charge density. Energy is the standard Coulomb functional, minimised over the wave functions  $\{\psi_i\}$ , the domain decomposition  $\{\Omega_i\}$ , and the nuclear positions; molecular geometry, bonding, and dynamics emerge from the same variational principle. The  $3N$ -dimensional configuration-space step is not taken at all; the computational obstacle Dirac identified is dissolved by working in 3D throughout, at the cost of replacing antisymmetry with strict spatial non-overlap.

In this respect RealQM is a direct realisation of **Dirac’s chemistry-as-applied-physics position** (the 1929 quote above) executed in the real-3D-space picture Schrödinger originally envisaged. RealQM takes Dirac’s program seriously by retaining only the underlying physics — Coulomb interactions between electronic wave functions and nuclei — and discarding the chemistry-specific machinery (basis sets, exchange–correlation functionals, fitted force fields) that StdQM had to introduce in order to make the configuration-space equations tractable. Chemistry then re-emerges, bottom-up, from the variational principle alone.

This reformulation — RealQM in what follows — is not merely a numerical recasting of the standard problem. It is a different mathematical object with different scaling: complexity grows with the number of mesh points in 3D, not with the number of basis functions in  $N \times 3D$ . As a consequence, the entire framework can be implemented in roughly 8000 lines of JavaScript and WebGPU compute shaders, runs in a browser on a consumer laptop GPU, and supports interactive simulation of systems with hundreds of explicit electrons in real time.

We organize the model as a hierarchy of reductions. **Level 1** is parameter-free: an atom is described as a nucleus of charge  $Z$  surrounded by  $N$  one-electron wave functions  $\psi_i$  on non-overlapping shell-shaped domains  $\Omega_i$ , with the configuration determined by minimum-energy packing. The Level-1 atomic model reproduces observed atom energies for elements Li through Rn to approximately 1% with no fitted constants. **Level 2** spherically homogenizes inner shells, keeping the shell occupancy intact but replacing the angular structure of  $\sum_{i \in \text{shell}} \psi_i^2$  with a radial density of matching total charge. **Level 3** further replaces the nucleus and inner shells together by a *pseudo-kernel* characterized by an effective charge  $Z_{\text{kernel}}$  and a softening radius  $r_c$ , leaving only the outermost valence wave functions explicit. **Level 4** combines Level-3 atoms into molecular assemblies. Two further levels extend the hierarchy beyond the wave-function

formalism: **Level 5** extracts a per-species reduced-model record (envelope, surface electrostatic potential, hydrophobicity, normal modes) from a converged Level-3 RealQM run on a single protein, with the wave function integrated out into geometric and electrostatic features; and **Level 6** drives a population of Level-5 records under Brownian dynamics, capturing  $\sim 10^6$ -protein cell-scale dynamics on the same hardware that runs Level-1 atoms. Each level is determined by comparison with the prior level, in the same architectural spirit as pseudopotentials and frozen-core methods in standard quantum chemistry but with a parameter-free ab initio bottom rather than empirical fits, and extending naturally upward through reduced-protein records to cell-scale population dynamics.

A second contribution of this paper concerns how the implementation was produced. The mathematical reformulation is the work of one author. The validation suite, the WebGPU compute shaders, the systematic kernel-architecture sweeps reported in Section 4, and the curated public Gallery were developed through extended interactive collaboration with **Claude**, a large-language-model AI assistant produced by Anthropic. The pattern of the collaboration was iterative and recorded: the human proposed mathematical and chemical questions; the AI translated them into runnable simulations, identified bugs, ran systematic parameter sweeps, and challenged claims that did not survive scrutiny; the human evaluated the results and corrected the AI’s interpretations when they slipped. The history of corrections — claims made, retracted, refined, restored — is itself part of the public record, encoded in the Gallery’s “Assessment by Claude” card.

We take this case to be a worth-reporting example of how a single mathematical mind, working with an AI engineering partner in real time, can produce research-grade interactive scientific software at a scale otherwise requiring a programming team. The collaboration is not the science, but it is part of how the science came into existence, and we report it as such.

**Position relative to standard quantum mechanics.** This paper does not present RealQM as a parallel approach to be weighed against StdQM. The position taken is that the configuration-space step of 1926–1929 was a wrong turn, that StdQM in its various interpretive guises (Copenhagen, Bohmian, Many-Worlds, decoherence-based) is the apparatus that grew up around that wrong turn, and that the foundational debate among those interpretations is a symptom of the wrong formulation rather than a genuinely open question about nature. The numerical successes of StdQM are real and document that the underlying physics — Coulomb interactions, the original real-space Schrödinger equation — is right; what those successes do not establish is that the  $3N$ -dimensional probability-amplitude reading is the correct foundation, and the Kohn 1998 observation makes this point from inside the discipline. RealQM is offered as the formulation Schrödinger should have arrived at in 1926 once the multi-electron case was admitted: a wave function on physical three-dimensional space, electrons as spatially-extended non-overlapping regions of charge density, a deterministic variational principle, and no probabilistic interpretation. The validation reported in Section 4 establishes that this formulation recovers the chemistry at the same numerical level as StdQM across atoms, molecules, reactions, and condensed phases; it does so at orders-of-magnitude lower computational cost; and it does so without the foundational difficulties that a century of work on the standard formulation has not resolved.

The paper is organized as follows. Section 2 states the reformulation and the hierarchical reduction. Section 3 describes the numerical implementation. Section 4 reports validation, including atom energies, the covalent  $\text{H}_2$  bond and its He limit, atomization energies of closed-shell hydrides across periodic-table groups, S66 dimer geometries, chemical reactions, protein folding, excited states, and a three-system condensed-phase calibration ( $\text{NaCl}$  crystal melt,  $\text{CO}_2$  dry-ice cluster, solid  $\text{N}_2$ ) that quantifies how the simulation’s overshoot factor against experimental transition temperatures scales with the missing-dispersion contribution. Section 6 states honest limits of the approach. Section 7 shows that stability of matter — the lower bound  $E \geq -CNZ^2$  on the ground-state energy of a system of  $N$  atoms — follows directly

from the Hardy inequality applied to non-overlapping electronic territories, in contrast to the heroic Dyson–Lenard–Lieb–Thirring proof required in standard quantum mechanics. Section 8 reports an analogous model of the atomic nucleus, demonstrating that the framework extends beyond chemistry to nuclear-scale physics. Section 9 places the work in the long-standing chemistry-vs-physics debate. Section 13 discusses the human–AI collaboration. The full source code, validation suite, and interactive Gallery are available at the URLs in Section 14.

## 2 RealQM Schrödinger Equation

### 2.1 Multiphase 3D formulation

The starting point is Schrödinger’s 1926 model of the hydrogen atom: a single real-valued wave function  $\psi : \mathbb{R}^3 \rightarrow \mathbb{R}$ , normalised by  $\int_{\mathbb{R}^3} \psi^2(\mathbf{x}) d\mathbf{x} = 1$ , with  $\psi^2(\mathbf{x})$  representing the electron charge density and the total energy

$$E(\psi) = \frac{1}{2} \int_{\mathbb{R}^3} |\nabla \psi(\mathbf{x})|^2 d\mathbf{x} - \int_{\mathbb{R}^3} \frac{\psi^2(\mathbf{x})}{|\mathbf{x}|} d\mathbf{x} \quad (1)$$

balancing kinetic localisation cost against Coulomb attraction to a unit positive kernel at the origin. The ground state minimises  $E(\psi)$  and gives the well-known  $E = -1/2$  Ha eigenvalue, in agreement with observation.

For an atomic system with  $N > 1$  electrons the standard approach (Schrödinger 1926, with the non-physical configuration-space step taken on Born–Heisenberg suggestion) is to extend  $\psi$  to a wave function  $\Psi(\mathbf{x}_1, \dots, \mathbf{x}_N)$  on  $\mathbb{R}^{3N}$ , with  $\Psi^2$  given a probabilistic interpretation. This step replaces three-dimensional physical space with an  $N$ -fold tensor product whose computational cost grows exponentially in  $N$  and whose physical meaning has been contested ever since.

**The RealQM formulation** keeps the wave function in physical  $\mathbb{R}^3$ . The  $N$ -electron wave function  $\Psi \in H^1(\mathbb{R}^3)$  is a sum of one-electron wave functions  $\psi_i$  on non-overlapping spatial domains  $\Omega_i$ :

$$\Psi(\mathbf{x}) = \sum_{i=1}^N \psi_i(\mathbf{x}), \quad \psi_i \in H^1(\Omega_i), \quad \int_{\Omega_i} \psi_i^2(\mathbf{x}) d\mathbf{x} = 1, \quad (2)$$

where  $\{\Omega_i\}_{i=1}^N$  is a partition of physical space into open non-overlapping regions,  $\Omega_i \cap \Omega_j = \emptyset$  for  $i \neq j$ , and  $H^1(\Omega)$  denotes the space of real-valued functions on  $\Omega$  with square-integrable first derivatives. The boundaries  $\Gamma_i = \partial\Omega_i$  are not specified in advance: they emerge as a free boundary, with the  $\psi_i$  being continuous across joint boundaries  $\Gamma_i \cap \Gamma_j$  as imposed by  $\Psi \in H^1(\mathbb{R}^3)$ , and the boundary location determined by the energy minimisation.

The total energy of  $\Psi$  is the sum of three Coulomb contributions:

$$TE(\Psi) = K(\Psi) + PK(\Psi) + PE(\Psi), \quad (3)$$

with kinetic-energy term

$$K(\Psi) = \frac{1}{2} \int_{\mathbb{R}^3} |\nabla \Psi(\mathbf{x})|^2 d\mathbf{x} = \sum_{i=1}^N \frac{1}{2} \int_{\Omega_i} |\nabla \psi_i(\mathbf{x})|^2 d\mathbf{x}, \quad (4)$$

attractive kernel-electron potential energy

$$PK(\Psi) = \int_{\mathbb{R}^3} W(\mathbf{x}) \Psi^2(\mathbf{x}) d\mathbf{x}, \quad W(\mathbf{x}) = - \sum_{a=1}^M \frac{Z_a}{|\mathbf{x} - \mathbf{R}_a|}, \quad (5)$$

and inter-electron repulsive potential energy without self-repulsion

$$PE(\Psi) = \sum_{i=1}^N \int_{\Omega_i} \sum_{j \neq i} V_j(\mathbf{x}) \psi_i^2(\mathbf{x}) d\mathbf{x}, \quad V_j(\mathbf{x}) = \int_{\Omega_j} \frac{\psi_j^2(\mathbf{y})}{2|\mathbf{x} - \mathbf{y}|} d\mathbf{y}. \quad (6)$$

The exclusion  $j \neq i$  in (6) removes self-repulsion by construction, since the regions  $\Omega_i, \Omega_j$  are disjoint and each electron contributes its own charge density only to the potentials seen by the other electrons. The nuclear–nuclear Coulomb term  $V_{NN}$  is included additively when nuclear positions are varied.

The ground-state configuration is found by minimising (3) over  $\Psi = \sum_i \psi_i$  with  $\psi_i \in H^1(\Omega_i)$ ,  $\int_{\Omega_i} \psi_i^2 = 1$ , and over both the partition  $\{\Omega_i\}$  and the nuclear positions  $\{\mathbf{R}_a\}$ . Variation with respect to  $\psi_i$  at fixed  $\Omega_i$  gives, with Lagrange multiplier  $E_i$  for the unit-norm constraint,

$$\mathcal{H}_i \psi_i \equiv \left( -\frac{1}{2} \Delta + W + 2 \sum_{j \neq i} V_j \right) \psi_i = E_i \psi_i \quad \text{in } \Omega_i, \quad (7)$$

$$\frac{\partial \psi_i}{\partial n} = 0 \quad \text{on } \Gamma_i, \quad (8)$$

the homogeneous Neumann condition arising from free variation of  $\psi_i$  over  $\Omega_i$ . Continuity of  $\psi_i$  across the inter-electron boundary plus (8) together comprise the **Bernoulli free-boundary condition** discussed in Section 2.4.

### Remarks.

- *Real-valued wave function in physical space.* The  $\psi_i$  are real-valued functions of the same spatial coordinate  $\mathbf{x} \in \mathbb{R}^3$ , so  $\Psi$  inherits the dimensionality of physical space. Because the  $\psi_i$  have disjoint supports, the cost of representing  $\Psi$  on a  $K$ -point grid is  $O(K)$ , rather than  $O(K^N)$  as in the configuration-space formulation.
- *Pauli exclusion through non-overlap.* The constraint that distinct electrons occupy disjoint domains plays the role of the Pauli exclusion principle. Antisymmetry of a many-body wave function is replaced by spatial separation of one-body wave functions; the two requirements are not equivalent, but the variational consequences (no two electrons share a quantum state) are recovered.
- *No probabilistic interpretation needed.* Each  $\psi_i^2$  is the actual electronic charge density of electron  $i$  in its own domain  $\Omega_i$ . There is no need to invoke “probability of finding electron  $i$  at  $\mathbf{x}$ ” — the density is the physical charge distribution.
- *Connection to AIM.* RealQM can be viewed as taking Bader’s Atoms-in-Molecules partitioning [5] as the *primary* variable: rather than first computing  $\rho(\mathbf{x})$  and then partitioning, the partition itself  $\{\Omega_i, \psi_i\}$  minimises the Coulomb energy directly.
- *Kinetic energy as localisation cost / self-repulsion.* The term  $K(\Psi) = \frac{1}{2} \int |\nabla \Psi|^2$  is named “kinetic energy” by analogy with Schrödinger’s hydrogen problem, but no electronic motion is implied. Two equivalent physical readings are available within RealQM. First, as a **localisation cost** that penalises rapid spatial variation of the charge density — a strain-like elastic energy of the electronic continuum that prevents collapse to a delta. Second, as the **self-repulsion** of each one-electron density  $\psi_i^2$  on its own domain  $\Omega_i$ : the gradient term acts as an internal pressure resisting compression of the charge cloud, in direct analogy with the bulk modulus of a soft continuum. The two readings are mathematically the same term; they differ only in physical interpretation. The self-repulsion reading makes transparent why the inter-electron Coulomb integral (6) correctly omits the diagonal  $j = i$ : that piece is already supplied, in a different mathematical form, by the gradient term. Standard one-body DFT must add a non-trivial self-interaction correction precisely because the analogous self-Coulomb piece is included in the Hartree term and then must be subtracted; RealQM’s disjoint-domain formulation removes the need for this correction by construction.



## 2.2 Comparison with the standard formulation

The standard formulation (StdQM) seeks an antisymmetric  $N$ -electron wave function  $\Psi^{\text{std}}(\mathbf{x}_1, \dots, \mathbf{x}_N)$  on  $\mathbb{R}^{3N}$  as an eigenfunction of the many-body Hamiltonian. Approximate methods truncate the variational space (HF: single Slater determinant; CC and CI: configuration expansions; DFT: a one-body density) [3, 4]. The exponential scaling of the configuration space is the underlying obstacle, and the probabilistic interpretation of  $|\Psi^{\text{std}}|^2$  replaces the direct charge-density meaning of  $\psi^2$  in Schrödinger’s  $H$ -atom (1).

The RealQM formulation (2)–(6) works in real  $\mathbb{R}^3$  at the level of  $N$  one-electron wave functions  $\psi_i$  on disjoint spatial domains, with antisymmetry replaced by strict spatial non-overlap. The two requirements are not equivalent — antisymmetric wave functions on  $\mathbb{R}^{3N}$  need not project onto non-overlapping one-body densities on  $\mathbb{R}^3$ , and non-overlapping unit-density wave functions do not span the antisymmetric subspace. The reformulation is therefore a different mathematical object, not a numerical approximation of the standard one. It is also closer in spirit to Schrödinger’s original 1926 picture of  $\psi^2$  as charge density than to the Born–Heisenberg configuration-space extension that produced StdQM.

This raises an obvious question: how do the two compare numerically? At the parameter-free Level-1 atomic model (Section 4.1) RealQM gives atom energies within  $\sim 1\%$  of observation. At higher levels (Section 4.4), Level-3 reductions match experimental atomization energies of closed-shell hydrides within 3–9% across multiple periodic-table groups. Where the methods part company is in the cost: a single closed-shell hydride at Level 3 runs in milliseconds on a laptop GPU, while CCSD(T) on the same system requires minutes to hours on a CPU.

Bader’s theory of *Atoms in Molecules* [5] partitions the total electron density into atomic basins via the topology of the gradient field. RealQM takes that partitioning idea seriously as the *primary* variable: rather than first computing  $\rho(\mathbf{x})$  from a Slater-determinant wave function and then partitioning, the partition  $\{\Omega_i, \psi_i\}$  minimises the Coulomb energy directly. This gives the AIM zero-flux partitioning the first-principles status that, in its original construction as post-processing on an externally-supplied density, it does not have.

## 2.3 The hierarchy of reductions

The Level-1 atomic model is parameter-free: the only input is the nuclear charge  $Z$ . As described in Section 4.1, shell structure ( $1s^2, 2s^2 2p^6, \dots$ ) emerges from minimum-energy packing rather than being imposed; electron sizes increase with distance from the kernel as a consequence of the variational principle, not as a pre-set rule. We use this model as the bottom of the hierarchy.

**Level 2** spherically homogenizes the inner shells: the angular structure is replaced by spherical symmetry, and the total charge of each inner shell is redistributed as a spherically symmetric radial density. Bond-relevant chemistry is not affected because inner shells contribute only to the effective Coulomb potential seen by valence electrons.

**Level 3** combines the nucleus and the spherically homogenized inner shells into a single *pseudo-kernel* with an effective charge  $Z_{\text{kernel}}$  and a softening radius  $r_c$ . The valence electrons remain explicit. The kernel is parameterized by  $Z_{\text{kernel}}$  and  $r_c$ ; both are determined by comparison with Level-2 results. This is the level used throughout Section 4.

**Level 4** combines Level-3 atoms into molecular assemblies. The valence electrons may be either single-orbital (no split) or split into angular sectors aligned with bond directions. The choice of splitting topology — sphere, hemisphere, third ( $120^\circ$ ), tetra ( $109.5^\circ$ ) — and the kernel charge  $Z$  together constitute the *architecture* of the model for that molecule.

Levels 1–4 are reductions *within* the RealQM formalism: each step simplifies the atomic-level description while keeping the wave function explicit. Two further levels are downstream applications that consume RealQM data rather than refine it, and the reader should note the conceptual break — at Level 5 the wave function is integrated out and no longer present at runtime.

**Level 5** extracts a *reduced-model* record from a converged Level-3 (or Level-4) RealQM run on a single protein species. The record contains the solvent-accessible isosurface, surface electrostatic potential, hydrophobicity map, net charge, hydrodynamic radius, diffusion coefficient, and the first  $\sim 10$ – $20$  functional normal modes from the  $C\alpha$  Hessian. One Level-5 record per species, of order 1–10 KB. The wave function is no longer present in the record; it has been integrated out into geometric and electrostatic features.

**Level 6** drives a population of Level-5 records under Brownian dynamics, capturing  $\sim 10^6$ -protein cell-scale dynamics on the same hardware that runs Level-1 atoms. Each species in the population is a separate Level-5 record; pairwise interactions between copies are mediated by the surface electrostatic and steric data in the records, with type-specific affinity terms for binding partners. Levels 5 and 6 thus realise the multi-scale bridge from atoms to cells outlined in Section 9.

We emphasize that this hierarchy is principled, not heuristic. Each level is derived from the prior level by a specific reduction (Levels 1–4 within the wave-function formalism; Levels 5–6 by integrating it out). The empirical content lies in choosing the right architecture for a given system at the appropriate level.

## 2.4 The free boundary and Bernoulli conditions

The non-overlap constraint  $\Omega_i \cap \Omega_j = \emptyset$  means that each one-electron wave function  $\psi_i$  is supported on its own spatial region  $\Omega_i$ , with the regions  $\{\Omega_i\}$  partitioning the relevant volume. The interfaces  $\Gamma_i \cap \Gamma_j$  between regions are not specified in advance — they emerge from the energy minimisation. The variational principle (3) is therefore a **free-boundary problem**, mathematically analogous to Bernoulli’s classical free-boundary problem in potential flow and to the Stefan problem for phase transitions.

At equilibrium, the boundary satisfies a **Bernoulli condition**:  $\psi_i$  is continuous from within  $\Omega_i$  toward the interface (no singular behaviour at the boundary), and the **normal derivative**  $\partial\psi_i/\partial n$  **vanishes** at the interface (homogeneous Neumann). The Neumann condition is not externally imposed; it emerges from the variational principle. When (3) is minimised jointly over the wave functions and the boundary location, the transversality condition at the boundary — vanishing first variation for arbitrary admissible boundary deformations — yields (8), i.e.  $\partial\psi_i/\partial n = 0$  on  $\Gamma_i$ .

Across the interface, the density  $\psi_i^2$  is positive on the  $\Omega_i$  side and zero on the other side, with the jump occurring sharply at  $\Gamma_i$ . The boundary condition is therefore the combination of (a) continuity of  $\psi_i$  as  $\Gamma_i$  is approached from within  $\Omega_i$ , (b) the homogeneous Neumann condition  $\partial\psi_i/\partial n = 0$  on  $\Gamma_i$ , and (c) a jump from  $\psi_i \neq 0$  inside to  $\psi_i = 0$  outside. We collectively refer to this as the Bernoulli condition by analogy with Bernoulli’s free-boundary problem.

When the system is **not at equilibrium**, density mismatches at the interfaces generate forces that drive boundary motion. The free boundary moves toward configurations satisfying the Bernoulli condition; the bulk wave functions and the boundary location relax together. This dynamic is part of how the variational principle arrives at the equilibrium configuration: the boundaries are not fixed but evolve as the wave functions relax. In the numerical implementation (Section 3), the relaxation is realised through a smoothed  $w$ -field that softens the partition during imaginary-time propagation and tightens to the Bernoulli condition at convergence.

The free-boundary perspective places the wave-function formulation in a well-studied class of variational problems for which regularity theory, level-set methods, and Dirichlet–Neumann decompositions are available.

**Equivalence with Bader’s zero-flux surfaces.** The Neumann condition (8) is the wave-function-level version of the **zero-flux condition** that defines atomic basins in Bader’s Atoms-in-Molecules theory [5]. There, atomic basins  $\Omega_i$  are the regions enclosed by surfaces  $\Gamma_i$  on which



the gradient of the total density vanishes in the normal direction:

$$\nabla \rho(\mathbf{x}) \cdot \mathbf{n} = 0 \quad \text{on } \Gamma_i. \quad (9)$$

In RealQM, the local density inside  $\Omega_i$  is  $\rho_i(\mathbf{x}) = \psi_i^2(\mathbf{x})$ , so  $\nabla \rho_i \cdot \mathbf{n} = 2\psi_i(\partial\psi_i/\partial n)$ . The variational Neumann condition (8) therefore enforces (9) automatically. The two characterisations — Bader’s topological zero-flux surface and the RealQM Bernoulli free boundary — describe the same geometric object, but with the directions of dependence reversed: in AIM the partition is computed *after* the density is fixed by an external wave-function method, while in RealQM the partition is a *primary* variational degree of freedom co-determined with the  $\psi_i$ . The Bernoulli condition thus gives the AIM zero-flux partitioning a first-principles status it does not have in its original construction.

The Bernoulli condition itself appears to be **difficult to verify or disprove by direct experimental observation**: the inter-electron interface is not a quantity that current experiments resolve, and standard quantum-chemistry methods do not produce a comparable object (they work with overlapping orbitals or a single total density, not a partition of space into electron territories with a sharp boundary). In computations, however, the condition functions as a working model: imposing continuity and homogeneous Neumann at the free boundary is consistent with the variational principle, the relaxation reaches a stable boundary configuration, and the resulting energies and geometries agree with experiment across the cases reported in Section 4. We therefore adopt the Bernoulli condition as a model assumption supported by computational evidence rather than by direct measurement, and note this status explicitly.

## 2.5 Forces from total electronic potential

In standard quantum chemistry, forces on nuclei are typically computed as derivatives of the total energy with respect to nuclear coordinates:  $\mathbf{F}_a = -\partial E / \partial \mathbf{R}_a$ . The Hellmann–Feynman theorem states that for an exact wavefunction this equals the electrostatic force from the electron density on each nucleus, but for approximate wavefunctions the two differ by Pulay terms arising from the basis-set dependence on nuclear positions. The energy-derivative approach is the standard way forces enter molecular dynamics, geometry optimization, and reaction-path calculations.

In the multiphase electron-density formulation, forces are computed **directly from the gradient of the total electronic potential**:

$$\mathbf{F}_a = -Z_a \nabla P(\mathbf{r})|_{\mathbf{r}=\mathbf{R}_a} \quad (10)$$

where  $P(\mathbf{r})$  is the total Coulomb potential generated by all electron densities and other nuclei, evaluated at the position of nucleus  $a$ . There is no energy functional being differentiated; the force is the local electrostatic gradient on each nucleus, computed directly from the electron density and other nuclear charges via Coulomb’s law.

The two approaches give numerically the same result at exact electron equilibrium (Hellmann–Feynman) but differ in conceptual standing. The local-potential view is closer to the physics: **nature does not carry a record of energy**, only of local potential gradients. A nucleus does not “know” the total energy of the molecule and then take a derivative; it experiences the Coulomb force from the surrounding charge distribution, locally and instantaneously. Dynamics in RealQM is therefore driven by physical forces evaluated pointwise from the density, not by differentiation of a global functional.

This has practical consequences. Energy-derivative forces in standard QC require careful treatment of basis-set dependences (Pulay forces), analytic-gradient code that scales as the wavefunction calculation itself, and consistency between the kinetic and potential terms in the energy functional. Local-potential forces in RealQM are evaluated directly on the real-space grid, are  $O(N)$  in the number of grid points, and require no additional gradient machinery beyond the density and nuclear positions already maintained. The simplification is structural:

by working with the electronic potential as the primary object driving dynamics, the framework avoids the energy-derivative bookkeeping that constitutes a substantial part of mainstream quantum-chemistry implementations.

### 3 Numerical implementation

The reformulation is implemented as `mol_fast.js` (~1600 lines) and `molecule.js` (~5300 lines), JavaScript modules that compile WGSL compute shaders to a WebGPU device. `mol_fast.js` uses unit-density orbitals with explicit angular splitting; `molecule.js` uses a Voronoi-partition labelling field. Both run entirely in the browser.

#### 3.1 The 3-line code: essence of the algorithm

The full RealQM solver, despite the size of the implementation, has at its mathematical core just three coupled time-evolution equations applied at every grid point and every time step. For  $i = 1, \dots, N$ :

$$\dot{\psi}_i + \mathcal{H}_i \psi_i = 0 \quad \text{in } \Omega_i, \quad \frac{\partial \psi_i}{\partial n} = 0 \quad \text{on } \Gamma_i, \quad (11)$$

$$\dot{V}_i - \Delta V_i = 2\pi \psi_i^2 \quad \text{in } \mathbb{R}^3, \quad (12)$$

$$\dot{\chi}_i + \beta_i |\nabla \chi_i| = 0 \quad \text{in } \mathbb{R}^3, \quad (13)$$

with  $\mathcal{H}_i$  from (7) and  $\psi_i$  renormalised to  $\int \psi_i^2 = 1$  at every step.

The three equations correspond to the three physical processes the framework needs:

- (11) is **imaginary-time propagation** of the one-electron Schrödinger-like equation: each  $\psi_i$  relaxes toward the eigenfunction of  $\mathcal{H}_i$  with the lowest eigenvalue, subject to the Bernoulli free-boundary condition. The non-self-consistent piece of  $\mathcal{H}_i$  depends only on the kernel potential  $W$  and on the instantaneous Hartree potentials  $V_j$  from the *other* electrons.
- (12) is the **Poisson equation** (in parabolic-relaxation form) for each one-electron Hartree potential  $V_i$ , sourced by  $\psi_i^2$ . Self-repulsion is excluded by construction because (11) sums  $V_j$  only over  $j \neq i$ .
- (13) is a **level-set evolution** of the indicator  $\chi_i$  of  $\Omega_i$ . The speed  $\beta_i$  at the inter-electron interface is set by the local jump of  $\psi_i^2$  across the boundary, moving the front toward configurations satisfying the Bernoulli condition (continuity of  $\psi_i$  at  $\Gamma_i$ ). At convergence,  $\dot{\chi}_i = 0$  and the partition is stationary.

Explicit time-stepping of (11)–(13) on a fixed real-space grid, with  $\psi_i$  renormalised at every step, constitutes the entire solver. Each line is one PDE that becomes one stencil-update kernel on the GPU; everything else (forces on nuclei, dipole moments, Lindemann diagnostics, kernel softening, angular splitting) is post-processing or initialisation. We refer to this as the *three-line code*: the framework’s substantive content fits in three lines, and the rest of the implementation is data plumbing around those three lines.

#### 3.2 Real-space grid

The simulation domain is a cubic box of side  $L$  a.u., discretized as  $N \times N \times N$  grid points (typical  $N = 100$ – $200$ ,  $L = 10$ – $22$  a.u., grid spacing  $h = L/N \approx 0.1$  a.u.). Each electron’s density is represented as an  $N^3$  array of floats (orbital amplitudes), evolved by imaginary-time propagation (ITP) of the Hamiltonian operator  $H = T + V$  on the grid. The Hartree potential  $P$  from each density is solved by parallel Poisson diffusion. Self-interaction is removed by subtracting the contribution of each electron’s own density from the total Hartree potential.

A typical step does  $\sim 10$  GPU compute dispatches: orbital ITP update, Poisson update, kinetic-energy reduction, normalization, force computation. On a consumer laptop GPU ( $\sim 10$  TFLOPS), this runs at 30+ steps per second for the systems reported in Section 4.

### 3.3 Kernel softening and angular splitting

A Level-3 kernel is parameterized by  $(Z_{\text{kernel}}, r_c, \text{split\_type}, \text{split\_idx}, \text{split\_axis})$ . The kernel potential at distance  $r$  from the nucleus is  $V_{\text{kernel}}(r) = -Z_{\text{kernel}}/r$  for  $r > r_c$ , with a smooth softening for  $r \leq r_c$  that matches the Coulomb tail and goes to zero at the origin. The `split_type` is one of {sphere, hemi, third, tetra, hemi\_third}, defining how the angular sectors of the kernel partition the orbital domain. The `split_axis` is the axis around which sectors are arranged. The `split_idx` selects which sector this particular orbital occupies.

For a heavy atom with multiple valence electrons, several sub-orbitals at the same nuclear position with different `split_idx` values together represent the full valence shell. Each sub-orbital evolves independently subject to its angular sector mask.

### 3.4 Code size and computational cost

The full RealQM implementation is approximately 8000 lines of JavaScript and WGS� combined. For comparison, mainstream quantum chemistry packages range from  $\sim 200,000$  lines (Quantum ESPRESSO) to  $\sim 3,000,000$  lines (Gaussian, NWChem). The size compression is enabled by working directly in real space: there are no basis sets, no two-electron integrals, no orbital coefficient bookkeeping, no analytic gradient machinery.

For typical jobs, RealQM runs orders of magnitude faster than CPU-based quantum chemistry on equivalent hardware: a water-dimer geometry optimization that takes hours of CCSD(T)/CBS computation completes interactively in seconds. A 216-water cluster with explicit electrons runs at real-time interactive frame rates on a single GPU; the equivalent DFT-MD simulation requires tens to hundreds of CPU cores running for weeks. The speedup factor of  $10^2$ – $10^4$  is real and is the practical breakthrough of the framework, even if accuracy at very high precision (CCSD(T)-class) is not the goal.

## 4 Validation

### 4.1 Level-1 atom energies, Li through Rn

We first validate the parameter-free Level-1 atomic model against observed atom total energies. The model is: a nucleus of charge  $Z$  surrounded by  $N$  unit electron densities, each occupying a non-overlapping spatial region, with the configuration that minimizes the total Coulomb energy taken as the ground state.

A central feature of the Level-1 model is that **shell structure is not preset. A shell structure partially conforming with the standard s, p, d organization emerges** from the variational principle as a consequence of minimum-energy packing: as electrons are added one at a time, each new electron occupies the lowest-energy spatial region available given the non-overlap constraint, and electron sizes naturally grow with distance from the kernel. The radial layering (a 2-electron inner shell, an 8-electron next shell, an 18-electron third shell, ...) corresponds to the standard  $1s^2$ ,  $2s^22p^6$ ,  $3s^23p^63d^{10}$  counts and is reproduced quantitatively in the energetics. The angular structure — the s/p/d distinction within a shell (spherical s, dumbbell p, four-lobed d) — is **only partially captured** by the unit-density partition, which divides each shell into spatial territories whose detailed angular shapes need not match the standard hydrogenic orbitals. The agreement of Level-1 atom energies with observation across Li–Rn is therefore evidence that the **radial** shell organization is correctly derived from packing, while the finer angular labelling of subshells remains an approximation in this Level-1 description.

The model has no parameters beyond specification of kernel charges and computational mesh size, typically  $\sim 0.1$  a.u. for minimal but functional resolution. Computed total energies for Li–Rn agree with observation to approximately 1% across the range, with the largest deviations at very heavy atoms where relativistic corrections are not included. (Full table in the public Gallery; see Section 14.)

This level provides the anchor for all subsequent reductions: each higher-level model (Level 2, 3, 4) is parameterized by comparison with Level-1 atom energies and Level-2 spherical densities.

## 4.2 Periodic Table from electron packing

The Level-1 model builds up the Periodic Table by sequentially adding electrons to a nucleus of charge  $Z$ , each electron occupying its own non-overlapping spatial domain  $\Omega_i$  with continuity of the global  $\Psi$  across  $\Gamma_i \cap \Gamma_j$  and homogeneous Neumann at the free boundaries (Section 2.4). Each electron takes on a width determined by proximity to the kernel while competing for space with the others — a “balloon-packing” problem in which the kinetic-energy term  $K(\Psi)$  from (4) acts as a compression cost. The resulting shell structure is therefore an emergent property of minimum-energy packing, not a presupposed orbital ansatz; this contrasts with StdQM, where shell occupancies follow from Aufbau filling of the  $s$ ,  $p$ ,  $d$  orbitals derived from the  $H$  atom.

**Shell sequence.** RealQM packs electrons into an expanding sequence of non-overlapping spherical shells  $S_n$  centred at the kernel,  $n = 1, 2, 3, \dots$ . Each  $S_n$  is split azimuthally into two half-shells, and each half-shell is further partitioned angularly into  $n \times n$  electron domains, giving  $2n^2$  electrons per filled shell. This yields the familiar sequence 2, 8, 18, 32,  $\dots$  which is repeated in the Periodic Table periods 2, 8, 8, 18, 18, 32, 32. For  $Z > 1$  the innermost shell  $S_1$  already contains *two* half-shell electron densities meeting at a common plane through the kernel, a structure inherited from the Helium atom ( $Z = 2$ ). This emergent splitting of  $S_1$  into two non-overlapping half-spaces accounts for the “two-valuedness” which in StdQM has to be supplied externally as the Pauli exclusion principle: in RealQM the non-overlap constraint between distinct electron territories does the same work, and the Pauli principle is not separately needed.

**Sub-shell structure.** RealQM further suggests that filled 8-shells are radially divided into two 4-shells, giving filled-shell decompositions of the form  $(2) + (4+4) + (4+4) + \dots$ . Computed ground-state energies for atoms across the Periodic Table, with each sub-shell homogenised to spherical symmetry in the simplified Atom Simulator code, reproduce observed atomic total energies to within  $\sim 1\%$  for elements as heavy as Radon. Representative results (RealQM versus observation, in Hartree):

| Atom    | Shell structure                            | RealQM $E$ | Observed |
|---------|--------------------------------------------|------------|----------|
| Carbon  | $(2) + (2+2)$                              | −38.2      | −37.7    |
| Neon    | $(2) + (4+4)$                              | −132.4     | −128.5   |
| Argon   | $(2) + (4+4) + (4+4)$                      | −523       | −526     |
| Iron    | $(2) + (4+4) + (14) + (2)$                 | −1260      | −1272    |
| Krypton | $(2) + (4+4) + (18) + (4+4)$               | −2766      | −2788    |
| Xenon   | $(2) + (4+4) + (18) + (18) + (4+4)$        | −7355      | −7438    |
| Radon   | $(2) + (4+4) + (18) + (32) + (18) + (4+4)$ | −22800     | −23560   |

**Why sub-shells?** A diagnostic test illustrates why the sub-shell structure is preferred energetically. With Neon at the proper  $(2) + (4+4)$  structure the computed energy is −132 Ha, in good agreement with the observed −128.5 Ha. With Neon forced into a single 8-electron outer shell,  $(2) + (8)$  without the  $(4+4)$  subdivision, the computed energy drops to −153 Ha, far below observation. The mechanism is the homogeneous Neumann condition acting between adjacent

sub-shells: this condition widens each shell and raises its kinetic-energy cost, and the variational principle therefore prefers the radially split configuration that splits this cost across two narrower sub-shells. Sub-shell structure is thus an emergent prediction of the variational principle, not a free choice.

**Recovery of Lewis’s 1916 cubic atom.** For the second-row L-shell ( $n = 2$ ), the variational packing of eight non-overlapping unit-density electron territories around a  $+2$  closed inner shell produces an arrangement in which the eight L-shell domains occupy the eight octants of a Cartesian frame centred on the kernel — a geometric pattern dictated by minimisation of the kinetic-energy localisation cost (4) subject to non-overlap. This is, geometrically, the eight-cornered cubic arrangement of L-shell electrons proposed by G. N. Lewis in his 1916 paper “The Atom and the Molecule” [19], originally introduced as a chemical bookkeeping device for valence and the octet rule and dismissed as quantum-mechanically unfounded after the Schrödinger 1926 orbital picture took hold. RealQM recovers Lewis’s geometric L-shell directly from the variational principle, without external octet axioms. The eight-electron L-shell of Ne, the four-electron half-shell of C, and the valence directionality of the second period thus all sit on a single first-principles footing, with Lewis’s chemically-motivated 1916 picture vindicated as the natural geometric result of the multiphase packing.

### 4.3 Chemical bonding: the $\text{H}_2$ covalent bond and its Helium limit

The hydrogen molecule  $\text{H}_2$  is the simplest covalent bond — two protons sharing two electrons — and provides the cleanest test of how covalent bonding emerges from the unit-density formulation. In RealQM,  $\text{H}_2$  is described as two unit electron densities that do not overlap (Bernoulli partition), distributed around two  $\text{H}^+$  kernels at separation  $R$ . The two electrons occupy opposite halves of the inter-nuclear region (the natural minimum-energy partition for two electrons in a symmetric two-kernel field). Each electron is attracted to both nuclei and repelled by the other electron through Coulomb interactions only; there is no exchange term and no antisymmetric wavefunction.

At a separation  $R \approx 1.6$  a.u. close to the experimental bond distance (1.40 a.u.), the parameter-free Level-1 calculation — each hydrogen contributing one unit electron density on its side of the Bernoulli partition — gives a total energy  $E = -1.1785$  Ha against the Kolos–Wolniewicz exact value  $-1.1745$  Ha (0.3% deeper than exact). The corresponding binding energy is  $\Delta E_{\text{bind}} \approx -112$  kcal/mol versus the experimental dissociation energy  $-109$  kcal/mol — **within 3%, with no fitted parameters**. The bond emerges from the same variational principle that organizes the atomic shells (Section 4.1): the electrons reorganize their territories to minimize the total Coulomb energy, and at the equilibrium  $R$  the two-kernel two-electron configuration is energetically preferred over two isolated H atoms.

**Bonding mechanism.** The bond is established by **accumulation of electron density between the two kernels**, which lowers the total kernel-attraction potential energy by bringing density closer to both nuclei. In the unit-density formulation, the two non-overlapping electron territories meet at the free boundary between the kernels, where **both densities are non-zero** — neither is required to vanish at the interface. This non-vanishing meeting (the Bernoulli condition of Section 2.4: continuity plus homogeneous Neumann) is precisely what makes the inter-nuclear accumulation energetically favorable. If the densities instead had to decay to zero between the nuclei (as they do at infinity for an isolated atom), accumulating density in the inter-nuclear region would require steep gradients and a balancing kinetic-energy cost that would offset the potential-energy gain. The Bernoulli partition lets each density remain smooth across the interface with finite values on both sides, so the inter-nuclear region accumulates electrons at low kinetic-energy cost while the kernel-attraction potential energy is enhanced. Covalent bonding in RealQM is thus a direct consequence of the free-boundary structure: **meeting non-zero densities**

allow the kernel potential energy to decrease through inter-nuclear accumulation without a balancing increase in kinetic energy. This conclusion is consistent with the R udenberg–Kutzelnigg analysis of the chemical bond [6, 7], in which careful decomposition of the wavefunction shows that covalent bonding is primarily a kinetic-energy lowering driven by orbital interference; the unit-density formulation reaches the same qualitative conclusion through a different route, namely the free-boundary partition rather than orbital overlap.

**The Helium limit: zero kernel distance, no nuclear repulsion.** A revealing limit is  $R \rightarrow 0$ . As the two H nuclei approach, their nuclear–nuclear repulsion  $1/R$  diverges, so  $\text{H}_2$  has a finite equilibrium separation. But if the two  $\text{H}^+$  point kernels are *merged* into a single  $+2$  point kernel — equivalently  $R = 0$  with the charges combined — the system becomes Helium ( $Z = 2$ , two electrons), and there is no inter-nuclear repulsion to oppose the merger. The Bernoulli partition is now two unit densities that do not overlap, both bound to the inner shell of the  $+2$  kernel, with their interface bisecting the kernel. This is exactly the Level-1 description of the He ground state.

The covalent bond and the atomic shell are therefore not separate phenomena; they are the same minimum-energy electron-packing principle applied at different nuclear separations. The continuous family parameterized by  $R$  interpolates between He at  $R = 0$  (kernels merged, no repulsion), the  $\text{H}_2$  bond at  $R = R_{\text{eq}}$ , and two isolated H atoms at  $R \rightarrow \infty$ . Each configuration is a stationary point of the same energy functional. The unification clarifies why a single architecture handles atoms and molecules: bonding is what the variational principle does when more than one nucleus is present, with the inter-nuclear repulsion setting the equilibrium separation.

#### 4.4 Atomization energies of closed-shell hydrides

We test the Level-3 reduction across a series of closed-shell hydrides  $\text{H}_nX$ , varying the heavy atom  $X$  across periodic-table groups. For each system, the kernel is parameterized by an effective charge  $Z_{\text{kernel}}$  and a softening radius  $r_c$ . We compute the total energy at the experimental equilibrium  $X$ –H bond length  $R$  and at twice that distance, and report

$$\Delta E_{\text{bind}} = E(R) - E(2R)$$

as a proxy for the atomization energy.

**Group 1 (alkali hydrides).** Single-electron  $X^{+1}$  kernel paired with a single H gives a closed-shell two-electron molecule. Sweeping  $r_c$  traces the alkali series:

| $r_c$ (a.u.) | $\Delta E$ (kcal/mol) | Real molecule     | Match            |
|--------------|-----------------------|-------------------|------------------|
| 0.50         | −48                   | NaH (−47)         | <b>within 2%</b> |
| 0.70         | −42                   | KH ( $\sim -43$ ) | <b>within 2%</b> |

The  $r_c=0$  limit of this single-orbital model corresponds to a 2-electron-in-one-territory configuration, distinct from the proper covalent  $\text{H}_2$  (Section 4.3).

**Group 2 (alkaline-earth dihydrides).** Linear H– $X$ –H with a  $+2$  kernel split into two hemispheres along the molecular axis (each holding one electron):

| $r_c$ (a.u.) | $\Delta E$ (kcal/mol) | Real molecule                             | Match            |
|--------------|-----------------------|-------------------------------------------|------------------|
| 0.40         | −140                  | $\text{BeH}_2$ (−144)                     | <b>within 3%</b> |
| 0.50         | −129                  | between $\text{BeH}_2$ and $\text{MgH}_2$ | regime           |



**Group 14 (XH<sub>4</sub> tetrahedral).** Central +2 kernel with a single 2-electron orbital paired with four hydrogen atoms at tetrahedral positions. A single architecture sweeps the entire group-14 series via  $r_c$ :

| $r_c$ (a.u.) | $\Delta E$ (kcal/mol) | Real molecule           | Match            |
|--------------|-----------------------|-------------------------|------------------|
| 0.20         | -369                  | CH <sub>4</sub> (-396)  | <b>within 7%</b> |
| 0.40         | -348                  | SiH <sub>4</sub> (-320) | within 9%        |
| 0.70         | -272                  | GeH <sub>4</sub> (-281) | <b>within 3%</b> |

**Group 16 (bent H<sub>2</sub>X).** Water requires a different splitting topology, with the axis along the H–O–H bisector. Both H atoms occupy the same hemisphere, with the lone pair in the opposing hemisphere as a paired sub-orbital. At  $r_c = 0.7$ ,  $\Delta E_{\text{bind}} = -225$  kcal/mol versus the experimental H<sub>2</sub>O atomization -232 kcal/mol — **within 3%**.

#### 4.5 What $r_c$ encodes

Across all four working groups, varying  $r_c$  with a fixed architecture traces a periodic-table column. We interpret  $r_c$  as the **inner-shell absorption radius** in the Level-3 reduction: the radius at which inner-shell electrons are absorbed into the kernel core, leaving only the explicit valence outside. As  $r_c$  grows, the kernel becomes more diffuse, the effective valence sees a larger inner core, and the bonding becomes weaker — exactly as down a periodic-table column. This interpretation is consistent quantitatively across groups 1, 2, 14, and 16, with periodic-trend match within 3–9% per element.

#### 4.6 Geometric validation against S66

We additionally test geometric agreement against Hobza’s S66 benchmark of non-covalent dimers. RealQM cannot directly validate S66 binding energies — those are CCSD(T)/CBS values in the -1 to -7 kcal/mol range, well below the model’s absolute-energy noise floor ( $\sim 0.1$  Ha). But the equilibrium distances and angles can be compared directly. For five H-bonded dimers tested (water dimer, methanol dimer, methylamine dimer, methylamine–water, formamide dimer), distances agree with CCSD(T) reference to within 1–5%, and force-direction diagnostics confirm the basin of attraction is correctly identified.

#### 4.7 Chemical reactions: proton transfer

The local-potential force formulation (Section 2.5) makes RealQM a natural framework for chemical reactions. A reaction proceeds because the local Coulomb forces on each nucleus push it along a path that passes through bond-breaking and bond-making configurations; nothing is computed by reference to an energy surface, a transition-state search, or a barrier height. The forces are the same Coulomb gradients used for equilibrium geometry — only the initial configuration differs.

We illustrate this with **proton transfer**, the prototypical Brønsted acid–base reaction.

**HF + H<sub>2</sub>O → F<sup>-</sup> + H<sub>3</sub>O<sup>+</sup>.** The donor H of HF is placed at hydrogen-bonding contact from the O of H<sub>2</sub>O. The fluorine is modeled as a +3 kernel with four valence electrons split into half-spaces ( $r_c = 1.0$ ); the bare proton has no electrons; the oxygen lone pair is a +2 kernel with  $r_c = 0.8$ . As the simulation runs, the F<sup>-</sup> electron density repels the proton outward while the O lone pair attracts it inward. The proton transfers spontaneously: at convergence, H<sub>t</sub>–O = 0.99 Å (covalent in the newly formed H<sub>3</sub>O<sup>+</sup>), F–H<sub>t</sub> = 1.91 Å (released), with the F<sup>-</sup> ··· H<sub>3</sub>O<sup>+</sup> ion pair stabilized at 2.4 Å contact. No barrier search, no transition-state geometry, no pK<sub>a</sub> input — the forces find the product configuration directly.

**HCl + NH<sub>3</sub> → Cl<sup>-</sup> + NH<sub>4</sub><sup>+</sup>.** The same setup with chlorine (+1 kernel,  $r_c = 1.0$ ) and nitrogen (+3 kernel,  $r_c = 0.5$ ). The ammonia lone pair pulls the proton; the chloride density pushes it away. At convergence: H<sub>t</sub>-N = 0.97 Å (covalent in NH<sub>4</sub><sup>+</sup>), Cl-H<sub>t</sub> = 1.53 Å (released), Cl<sup>-</sup> ··· NH<sub>4</sub><sup>+</sup> contact 2.1 Å. The reaction completes in a single force-driven trajectory.

**Bond breaking and exchange: H + H<sub>2</sub> → H<sub>2</sub> + H.** A hydrogen atom approaches an H<sub>2</sub> molecule along the bond axis. RealQM follows the symmetric H<sub>3</sub> transition geometry directly via the local forces: the two H-H distances pass through equality at the saddle point and then bifurcate in the product channel, with the original bond broken and a new bond formed between the incoming H and the proximal hydrogen. The same Coulomb gradients that hold H<sub>2</sub> together also break it and re-form it.

**What this validates.** These cases test the **forces-not-energies** principle (Section 2.5) on systems where standard quantum chemistry would require either a transition-state search (for the saddle) or QM/MM molecular dynamics (for the trajectory). In RealQM the Coulomb forces that determine equilibrium geometry also determine reactive trajectories; there is no separate machinery for reactions. The chemical specificity — which proton transfers, in which direction, to what acceptor — emerges from the kernel parameters ( $Z$ ,  $r_c$ , split topology) of the participating atoms and is controlled by the local electron-density configuration around each nucleus, not by any reaction-specific input. Nature does not consult pK<sub>a</sub> tables or compute free-energy differences; it acts through local forces, and so does RealQM.

**Drug mechanisms as reactive scans.** The same forces-not-energies principle extends to a set of pharmacological mechanisms, run as *comparative reactive scans*: a generic nucleophile (NH<sub>3</sub>, an amine/thiol proxy) is driven toward an electrophilic carbon on an *activated* species and on a matched *control*, and the diagnostic is the differential drive =  $\Delta E(\text{control}) - \Delta E(\text{activated})$  as a function of approach distance, with a far reference removing long-range dipole bias. A positive, growing drive means the activating group genuinely lowers the reactive barrier. The framework reproduces the qualitative activation of several covalent-drug motifs: proton-pump inhibitor (omeprazole) activation by acid protonation;  $\beta$ -lactam antibiotic ring-strain activation (4-ring vs 5-ring control); covalent kinase-inhibitor warheads (acrylamide Michael acceptor vs plain alkene); and the pyridinium electron-sink of the PLP cofactor in GAD65 decarboxylation. A clear scope boundary appears: the perpendicular-approach scan captures *addition* mechanisms but fails for *substitution* (S<sub>N</sub>2, acyl transfer), where the reactive trajectory is backside and the leaving group departs as an anion — the reduced model’s known weak spots (anions, and a rigid scan geometry). These are qualitative sign/trend results on ~15–20-atom active-site cores; on such small mechanisms, established DFT/QM-MM is the more accurate tool, and the value here is the *parameter-free, browser-scale* recovery of the correct activation physics, not competitive barrier heights. All scans are public in the Gallery (Section 14).

## 4.8 Protein folding

The reformulation supports molecular dynamics at scale. We have used the framework to simulate folding of small peptides and mini-proteins driven by quantum-mechanical forces from the unit-density model, supplemented in some cases by a single implicit-solvent hydrophobic parameter. Headline results:

**Polyglycine 12-residue  $\beta$ -hairpin (dry).** Starting from an extended chain at 150° opening angle, the hairpin folds spontaneously to ~105° driven by interstrand N-H···O=C hydrogen bonds. No empirical force field is used; the H-bonds are quantum-mechanical, computed from the gradient of the electron-density potential at each backbone atom. Folding stalls at 105° where

H-bond attraction balances backbone repulsion. Adding explicit water does not improve folding — the entropic hydrophobic effect, which completes folding in real biology, is not captured by an electronic-energy solver.

**8-residue polyglycine alpha-helix from a near-linear start.** Starting from a chain with radius 0.5 Å and rise 3.0 Å per residue (essentially extended), the chain folds spontaneously into a helix of radius 2.1 Å and rise 1.63 Å per residue (within 9% of ideal alpha-helix geometry). The helix score reaches 67%; two of four characteristic  $i \rightarrow i+4$  hydrogen bonds form at H...O distances near 5 Å. The driving forces are the same quantum-mechanical H-bonds; a mild radial helical bias (coefficient 0.03) assists curling.

**Chignolin (10-residue mini-protein).** GYDPETGTWG folds from 135° to 40° opening angle when supplemented with a SASA-based implicit hydrophobic parameter ( $\gamma = 5.0$ ). Side-chain electron repulsion that would otherwise unfold a small protein in vacuum is overcome by the hydrophobic pressure on solvent-exposed surfaces.

**Villin HP35 (35-residue 3-helix bundle).** Under universal  $i \rightarrow i+4$  H-bond biases applied to every residue (with no specification of which segments are helical) plus the SASA hydrophobic parameter, 7 of 9 native helix H-bonds form at the right distances and the Phe core packs to a 5.6 Å contact distance against a 6.0 Å target. This is “near-blind” folding: no native contacts and no segment-specific helix specifications are used; only two generic biological rules ( $i \rightarrow i+4$  H-bonds favored throughout; hydrophobic surfaces compact).

**Architecture and computational cost.** Each backbone atom is modeled at Level 3 with kernel charges  $Z = 4$  (C), 3 (N), 2 (O), 1 (H), with bond and angle constraints holding the protein backbone. Folding trajectories run interactively on a  $200^3$  or  $300^3$  grid; a 35-residue protein folds in hours of real-time on a single GPU, compared to weeks of CPU-cluster time for full DFT-MD or specialized hardware (Anton [11]) for force-field MD on the same timescales. The combination of ab initio H-bond forces and a single hydrophobic parameter, with no empirical force field, distinguishes this approach from both classical MD (which requires fitted parameters for every interaction) and pure ML methods (which do not solve the electronic-structure problem at all).

## 4.9 Reduced-form docking: streptavidin–biotin recognition

The Forward look in Section 9 proposed the cell-scale architecture in which each protein species is compressed to a small Level-5 reduced record — surface points carrying position, charge, and hydrophobicity, extracted from a converged Level-3 RealQM run — and the cell-scale dynamics is Brownian motion of such records interacting through Coulomb, steric, and hydrophobic forces. We here test that the reduced record retains enough information for *recognition*: does an enclosed binding pocket, represented only by Level-5 data, discriminate ligands by chemistry the way the underlying protein does?

The test system is the biotin-binding pocket of streptavidin, one of the tightest non-covalent receptor–ligand pairs known ( $K_d \approx 10^{-15}$  M). The pocket is represented by fourteen residues from the literature site: four tryptophans (W79, W92, W108, W120) lining the hydrophobic walls; three floor donors (N23, S27, S45); interior hydrophobics (A86, V47); a rim aromatic (Y43); and the rim arginine R84 that anchors biotin’s valeryl carboxylate. Each residue is rendered as a sphere of surface points; hydrophobicity is restricted to the inward-facing hemisphere, reflecting the buried packing of the real sidechains. Four ligand types — a biotin-like model (hydrophobic body with an anionic tail), a neutral hydrophobic ball, a cationic ball, and an anionic ball — are docked by Brownian dynamics under identical Coulomb, steric, and hydrophobic potentials. Each ligand was run for 50 independent trials of 40 000 BD steps starting at random orientation

and random direction at 13 Å from the pocket centre, biased to the upper hemisphere so all ligands have access to the pocket opening. A trial counts as “in pocket” if the ligand centre of mass reaches within 6 Å of the pocket centre with  $\geq 6$  contact points and binding energy below  $-3$  (in the dimensionless units of the simulation).

| Ligand                   |                                   | bound | in pocket   | Mechanism                             |
|--------------------------|-----------------------------------|-------|-------------|---------------------------------------|
| biotin-like              | (hydrophobic head + anionic tail) | high  | 24/50 (48%) | hydrophobic burial + R84 tail anchor  |
| neutral hydrophobic ball |                                   | 25/50 | 20/50 (40%) | hydrophobic only, no orientation lock |
| anionic ball             |                                   | 50/50 | 12/50 (24%) | R84 captures at rim, no body to bury  |
| cationic ball            |                                   | 0/50  | 0/50 (0%)   | R84 electrostatically excludes        |

The hierarchy reproduces three independent recognition modes simultaneously, from a Level-5 record alone:

*Charge complementarity.* The cationic ligand is electrostatically excluded by R84 (0/50), while the anionic ligand is electrostatically captured at the rim (50/50). This is the sign-discriminating behaviour that any selective receptor must exhibit.

*Hydrophobic burial.* The neutral hydrophobic ball, lacking the anionic tail, finds the pocket purely by hydrophobic attraction and enters in 40% of trials; the cationic ball, despite carrying the same hydrophobic skeleton, never enters because R84 keeps it at distance. The 40% rate quantifies the unbiased burial probability of a featureless hydrophobic body once stochastic search is given enough time.

*Combined recognition.* The biotin-like ligand benefits from both effects: R84 pulls the anionic tail upward and orients the ligand head-down, after which the hydrophobic body falls into the well. The 48% pocket-entry rate is the highest in the series. The orientation lock is the same mechanism by which real streptavidin recognises biotin from a vast library of similar small molecules: an enclosed hydrophobic cavity is necessary but not sufficient; the specific anionic-tail-to-rim-arginine register selects *this* ligand.

**Pocket vs no-pocket control.** The same docking protocol applied to chignolin (10-residue  $\beta$ -hairpin, no enclosed cavity, surface-only sidechains) gives the complementary result. With a neutral hydrophobic ligand, only  $8/50 = 16\%$  of trials produce stable surface contact, against 40% pocket entry for streptavidin. With a cationic ligand, chignolin binds preferentially at the rim of the D2/E4 acidic patch (Y1 aromatic anchor,  $\sim 19/50$  surface contacts), while streptavidin excludes it entirely. The framework therefore discriminates not only by chemistry of the ligand but by topology of the receptor: an enclosed cavity yields high specific binding, an exposed surface yields low transient binding. The same Level-5 record reproduces this distinction with no parameter retraining — only the surface point distribution changes.

The result is independent confirmation that the Level-5 reduced record is not merely a coarse-grained shape but a functional surrogate for the underlying receptor. The same record that drives the cell-scale Brownian gas in the Forward look has, on its own, enough information to reproduce the three classical pillars of drug–receptor discrimination, together with the basic topological pocket-vs-surface distinction.

**Scope of the validation.** What the docking demonstrations validate is the *Level-5 to Brownian-dynamics* step: that a small surface-point record — per-residue position, sidechain charge, and inward-facing hydrophobicity — carries enough information for a deterministic force model (Coulomb + steric + hydrophobic) to reproduce the canonical recognition modes. The Level-5 records used here (chignolin, streptavidin, barnase, barstar) are constructed manually from literature data: PDB  $C_\alpha$  coordinates for the relevant residues, Kyte–Doolittle hydrophobicity assigned by amino-acid type, and standard sidechain formal charges. They are not the output of a converged RealQM Level-3 run on each of those proteins.

The complementary step — *Level-3 to Level-5* extraction from a converged RealQM electron density — is shown separately on chignolin in the Gallery, where surface electrostatic potential, hydrophobicity map, and solvent-accessible isosurface are read off the converged Level-3 state. Running the full Level-3 simulation on streptavidin or barnase before extracting their Level-5 records is computationally larger than the small-peptide cases in Section 4.8 and we have not done it here. The Forward look in Section 9 sets out that full chain — Level-3 quantum simulation per protein species, then Level-5 reduction, then cell-scale Brownian gas — as the architecture; the present subsection validates its terminal step and shows it has the discriminating power claimed for it.

**Why the hand-built records are a fair surrogate.** The chignolin extraction shows what a Level-5 record produced by RealQM looks like in practice. Three observations transfer to the streptavidin and barnase–barstar cases. *Charges*: integrated electron density around each sidechain at physiological conditions reproduces the standard formal charges (D/E carboxylate  $\rightarrow -1$ , K/R guanidinium/ammonium  $\rightarrow +1$ , H  $\rightarrow 0$  to  $+1$ ) that the hand-built records use. *Hydrophobicity*: the Kyte–Doolittle scale is an experimentally fitted summary of the same surface-density polarity that RealQM computes, and the two agree per residue to within tens of percent — well below the precision needed to discriminate the four ligand types in the table above. *Surface geometry*: RealQM would replace each fixed-radius sphere of points by the actual solvent-accessible isosurface of the converged density, anisotropic with respect to sidechain conformation, but with the same topology of inward-facing hydrophobic patches and outward-facing charged groups that controls the docking outcome. The recognition modes that the docking demonstrations resolve — charge complementarity, hydrophobic burial, R84 orientation lock, the long-range electrostatic steering of barnase–barstar — are set by the relative positions of charged and hydrophobic patches, not by their precise shape. The conclusion that the framework recognises pockets, anchors anionic tails, and steers oppositely charged faces into contact would therefore survive replacement of the hand-built records by RealQM-extracted ones.

#### 4.10 Protein–protein interactions and cell-scale population behaviour

The docking validations in Section 4.9 test a small ligand against a single receptor. The next question is whether the same Level-5 framework, applied to two proteins of comparable size, discriminates a specific protein–protein partner from non-partners, and whether that discrimination survives at population scale in a crowded environment.

**Barnase–barstar: electrostatic steering of a specific PPI.** The barnase–ribonuclease-inhibitor pair is the textbook case of long-range electrostatic steering in protein recognition, with  $K_d \approx 10^{-14}$  M. The Level-5 record places barnase’s basic interface (K27, R59, R83, R87) on its  $+x$  face and the catalytic Y103 nearby; barstar’s acidic interface (D35, D39, E76, E80) sits on its  $-x$  face with a tryptophan and threonine as polar partners. With charges concentrated on outward-facing hemispheres so each protein presents a real dipole moment toward its binding face, and with neutral body residues, the long-range monopole attraction  $(+4) \times (-4)$  pulls the partners together while the dipole alignment torque orients barstar’s red face to meet barnase’s blue face before short-range contact. The result is a clean native-interface lock-in in essentially every trial: 5 out of 5 trials reached the bound state in initial testing, with iface contacts  $> 5$  and binding energy below  $-3$  in the simulation’s units.

**Chignolin homodimer: mutual exclusion.** The same docking protocol applied to two identical chignolin records yields the opposite, equally meaningful result. With both partners carrying net charge  $-2$  (D2 and E4 acidic in each), Coulomb repulsion at any approach distance exceeds the weak attraction available from two small aromatic patches (Y1 and W8). The two

chignolins never associate stably — exactly as real chignolin behaves in solution, where it is a monomer at neutral pH rather than a homodimer. The framework therefore not only predicts specific binding when it should occur but also predicts non-binding when it should not. This is the negative-result side of selectivity that any model claiming to predict drug–receptor or PPI specificity must clear.

**From dimer to population.** With the two-protein limit established, the natural next step is to put many proteins in a periodic box and ask whether the specificity survives crowding. We use the mutual-closest pairing rule: each protein chooses its closest bound contact, and a unique pair is recorded only when both proteins choose each other, so each protein participates in at most one counted dimer. The random baseline for  $N+N$  proteins under this rule is  $N/(2N-1)$  BS fraction — approximately 50.5% for 50+50 and 52.6% for 10+10 — since each protein has  $N$  different-species and  $N-1$  same-species candidates at random.

A simulation of 10 barnase plus 10 barstar in a 60 Å periodic box, all proteins diffusing and rotating freely under the same Coulomb + steric + hydrophobic forces, yields 9 unique barnase–barstar dimers against 0 barnase–barnase and 0 barstar–barstar pairs in equilibrium. That is 100% specificity against a random expectation of 52.6% BS. Each protein finds its complementary partner across a crowded box of like-charged competitors.

Scaling to 50 barnase + 50 barstar in a 90 Å box gives 25 unique BS dimers against 1 BB and 0 SS pairs after 1500 BD steps — 96% specificity at population scale (random baseline 50.5%), with the remaining  $\sim 25$  proteins still searching in transient contact. The smaller 60 Å system reaches near-saturation; the larger box equilibrates with a lower fraction paired but the same near-perfect specificity.

| Setup                   | Box  | N proteins | uBS      | uBB   | uSS   |
|-------------------------|------|------------|----------|-------|-------|
| small soup, dense       | 60 Å | 10 + 10    | 9        | 0     | 0     |
| larger soup, dense      | 90 Å | 50 + 50    | 25       | 1     | 0     |
| random baseline (50+50) | —    | 50 + 50    | 50.5% BS | 24.7% | 24.7% |

The framework reproduces four distinct features of real cell biology in a single force model: *specific binding* when charge complementarity and hydrophobic burial align (barnase–barstar); *mutual exclusion* when like-charged partners face each other (chignolin–chignolin); *specificity under crowding* when many proteins of mixed species share a confined volume (PPI soup,  $\geq 96\%$  unique-pair BS specificity against a random baseline of  $\sim 50\%$ ); and *liquid–liquid phase separation* when multivalent sticker proteins concentrate at a critical density (hexavalent hydrophobic patches,  $\sim 60\%$  condensed fraction in dynamic coexistence with a dilute phase). The smallest non-trivial population of the cell-scale architecture — twenty proteins of two species — already exhibits the qualitative behaviour predicted by the Forward look in Section 9.

**Condensate formation: liquid–liquid phase separation.** A fundamental cell-biology phenomenon distinct from specific binding is the spontaneous segregation of certain proteins into liquid-like droplets that coexist with a dilute surrounding phase — membraneless condensates such as those formed by FUS, hnRNP and P-granule proteins. The thermodynamic driving force is multivalent weak interaction: each protein has multiple "sticker" patches that can engage multiple partners, producing a percolated network that prefers a dense phase over a dilute one. We model this in the same Level-5 framework by replacing the single-interface protein of the previous paragraphs with a *hexavalent sticker protein* carrying six hydrophobic aromatic patches (Y/W/F clusters) on the six  $\pm x, \pm y, \pm z$  faces. No charges — charge-charge same-sign repulsion would interfere with multivalent contact — only aromatic hydrophobic stacking, exactly the chemistry that drives real biological condensates.

Sixty hexavalent sticker proteins in a 70 Å periodic box at  $kT = 0.15$  in the simulation's units equilibrate to a dynamic two-phase state: a single dominant cluster of  $\sim 15$ –30 proteins



coexisting with smaller oligomers and dispersed monomers, with the condensed fraction (proteins in clusters of  $\geq 4$ ) fluctuating in the range 50–70%. The fluctuations themselves are biologically meaningful: a static aggregate would not show them, while a real droplet maintains continuous exchange between dense and dilute phases on a timescale faster than droplet identity. Switching to a neutral control protein with no sticker patches gives uniform dispersion at the same density and temperature. The framework therefore reproduces both the existence of liquid–liquid phase separation as a function of valency and the qualitative dynamics of the phenomenon, on a Level-5 record indistinguishable in structure from the records used for specific PPI in the previous paragraphs — only the sticker geometry differs.

**Competitive specificity with a decoy.** A stricter test of the framework’s discrimination is to add a third species to the soup — a competitor that resembles barstar in some respects but lacks the full native interface. We run 20 barnase + 20 barstar + 20 decoy in an 80 Å box. Three decoy variants probe a graded specificity gradient:

| Decoy variant                               | Decoy net charge | Barnase specificity uBS / (uBS+u |
|---------------------------------------------|------------------|----------------------------------|
| Same charge as barstar; W→S only            | −4               | 54%                              |
| Half charge; no hydrophobic core            | −2               | 60%                              |
| Neutral peptide; no charge, no hydrophobics | 0                | 86%                              |
| random baseline                             | —                | 50%                              |

The hierarchy reproduces the biological pattern that binding specificity is not all-or-nothing but a continuous response to interface quality. A single point mutation at the interface (W→S) drops the decoy’s competitive advantage modestly but does not eliminate it; halving the surface charge yields a stronger but still partial barstar preference; complete charge neutralisation yields the expected near-clean dominance. This is the same affinity gradient that single-residue mutagenesis exposes in real protein interfaces (Buckle, Schreiber and Fersht, 1994, and successors): every interface contact contributes a fraction of the total binding free energy, and removing any one weakens but rarely abolishes the interaction. The framework reproduces this property as a structural consequence of the force model, with no fitting.

**Implications for the cell-scale programme.** A bacterial cell is approximately  $10^6$  proteins of  $\sim 10^3$  species. The validation chain demonstrated here — atom → molecule → small-molecule docking → protein–protein dimer → population of mixed species → competitive specificity under decoy challenge — spans five of the six levels of the Forward-look hierarchy. The remaining gap is a quantitative scale-up: the framework as implemented here runs at  $\sim 10^2$  proteins in browser-JS,  $\sim 10^3$  on a single consumer GPU, and would reach  $\sim 10^6$  on a small GPU cluster with the same force model plus spatial hashing and long-range multipole expansions. The expensive part of the upstream pipeline — the Level-3 RealQM run that produces a Level-5 record for each species — is paid once per species and amortised across all copies at runtime. With approximately  $10^3$  species in a bacterial proteome, this is a tractable computational programme rather than a conceptual one.

**Honest scope: from caricature to functional cell.** The qualifier “cell simulation” covers two very different targets. A *caricature cell* — a periodic box containing every species of a bacterial proteome at correct copy number, diffusing and binding with experimentally meaningful kinetics — is a direct extension of the validation chain above and is reachable on a small GPU cluster within a year or two of focused engineering. The Level-3 RealQM runs that produce each Level-5 record can be parallelised across species; the Level-5 Brownian dynamics scales nearly linearly in protein count with spatial hashing and long-range multipole acceleration. Population-level observables — encounter rates, residence times, specificity under crowding, the

formation of membraneless condensates by phase separation, mutational effects on affinity — all sit on the existing pipeline.

A *functional cell* with metabolism, growth, division and signal transduction is further off and depends on extensions that the present formalism does not yet supply. *Conformational flexibility and allostery*: Level-5 records are rigid by construction, while real proteins change shape on binding, and signal transduction propagates through such changes. The natural extension — carrying the first  $\sim 10$  Hessian normal modes as additional internal degrees of freedom on each record — is clean conceptually but has not been implemented at scale. *Lipid membranes and the cytoskeleton*: the cell envelope and its internal scaffolding are continuum phase phenomena, not protein–protein binding, and require additional machinery beyond what the multiphase electron-domain formulation alone provides. *Reactive chemistry inside the cell*: enzymatic catalysis and metabolic pathways are intrinsically reactive, and RealQM does handle reactions at the atomic level (Section 4.7), but inserting reactive sub-steps inside a cell-scale Brownian gas is unsolved. *Water*: the hydrophobic effect appears here as a phenomenological  $1/r^7$  force tuned to reproduce binding outcomes, not as a thermodynamic free-energy difference derived from explicit solvent; for specific affinities this is borderline, for kinetics it is worse. *Nucleic acids*: DNA and RNA have charge densities and conformational regimes the protein-tuned Level-5 record does not yet capture cleanly.

The familiar analogy — “if we can simulate weather or traffic with millions of components, why not a cell?” — is weaker than it sounds. Weather and traffic do not need chemistry, recognition, or reactions; cells are exactly those things at the same component count. What the present framework offers is not a finished cell simulator but a credible *foundation*: a single self-consistent quantum source for the entire proteome, computed once per species, in place of the patchwork of DFT, CCSD, classical force fields and machine-learned potentials with inconsistent calibration regions that current cell-scale simulations have to stitch together. Whether the additional layers — flexibility, membranes, reactions, water, nucleic acids — prove tractable is an open question. What is no longer open is whether the foundational level is reachable in principle.

#### 4.11 Excited states: orthohelium and reduced-kernel atomic spectra

The unit-density framework extends naturally to excited states. The basic case is the orthohelium ( $1s\,2s$ ,  $^3S$ ) state of helium, in which two electrons occupy distinct spatial regions corresponding to the  $1s$  and  $2s$  shells. In RealQM, this is a directly accessible configuration of the Level-1 model: place two non-overlapping unit densities, the inner localized in the  $1s$  shell and the outer in the  $2s$  shell, and minimize the total Coulomb energy under the non-overlap constraint. The resulting configuration is an excited state of helium, distinct from the ground state ( $1s^2$ ,  $^1S$ ) where both electrons occupy the same shell.

The orthohelium case is included in the Gallery as a worked example. It illustrates that the framework is **not restricted to ground states**: any configuration of non-overlapping unit densities corresponding to a chosen shell occupancy is an admissible solution, and excited-state energies follow from the same variational principle applied to the chosen configuration. This is in contrast to standard QM where excited states require additional machinery (TDDFT, CIS, EOM-CC); here, they are just different choices of which shells the electrons occupy.

**Reduced-kernel atomic spectra.** The orthohelium picture generalises to a systematic test: take the Level-3 reduced kernel of charge  $Z_{\text{kernel}}$  and softening radius  $r_c$ , and ask whether it reproduces the *outer-electron spectrum* of an atom whose closed inner shells (and possibly an inner valence pair) have been absorbed into the kernel. The outer electron is then governed by the radial Schrödinger equation

$$-\frac{1}{2}u'' + \left[ \frac{\ell(\ell+1)}{2r^2} - \frac{Z_{\text{kernel}}}{r} \right] u = Eu, \quad r \in [r_c, R_{\text{max}}], \quad (14)$$

with empty-core ( $\psi$  excluded inside  $r_c$ ) and homogeneous Neumann boundary  $\psi'(r_c) = 0$  at the inner boundary, and  $u(R_{\max}) = 0$  at the outer. This is exactly the Level-3 reduced kernel of Section 2 restricted to its single explicit valence electron. We solve (14) by inward shooting plus bisection and fit ( $Z_{\text{kernel}}, r_c$ ) to NIST atomic spectra. Table 1 reports the result.

| Group          | Atom       | Configuration                            | $Z_{\text{kernel}}$ | $r_c$ (a.u.) | RMS (eV) |
|----------------|------------|------------------------------------------|---------------------|--------------|----------|
| 1 (alkalis)    | Li         | [He] + 2s                                | 1.00                | 1.95         | 0.05     |
|                | Na         | [Ne] + 3s                                | 0.95                | 2.00         | 0.11     |
|                | K          | [Ar] + 4s                                | 1.00                | 2.90         | 0.12     |
|                | Rb         | [Kr] + 5s                                | 1.00                | 3.10         | 0.14     |
|                | Cs         | [Xe] + 6s                                | 1.00                | 3.30         | 0.14     |
| He triplet     | He ortho   | 1s + outer (triplet)                     | 1.00                | 2.53         | 0.10     |
| He singlet     | He para    | 1s + outer (singlet)                     | 1.04                | 3.25         | 0.12     |
| 2 (alk. earth) | Be triplet | [He] + 2 valence ( $^3\text{L}$ )        | 1.30                | 4.36         | 1.1      |
|                | Be singlet | [He] + 2 valence ( $^1\text{L}$ )        | 1.00                | 3.36         | 0.11     |
|                | Mg triplet | [Ne] + 2 valence ( $^3\text{L}$ )        | 1.21                | 5.15         | 0.74     |
|                | Ca triplet | [Ar] + 2 valence ( $^3\text{L}$ )        | 1.58                | 6.25         | 0.27     |
| 13             | B          | [He] + 2s <sup>2</sup> + 2p              | 1.00                | 3.92         | 1.4      |
|                | Al         | [Ne] + 3s <sup>2</sup> + 3p              | 1.31                | 4.83         | 0.92     |
| 14             | C          | [He] + 2s <sup>2</sup> + 2p <sup>2</sup> | 1.51                | 3.19         | 2.2      |

Table 1: Reduced-kernel atomic spectra: ( $Z_{\text{kernel}}, r_c$ ) fitted to NIST outer-electron levels (s, p, d series,  $n$  up to 5–6) for the indicated atomic configurations.  $Z_{\text{kernel}}$  locks at  $\sim 1$  (residual after closed-shell screening) for the cleanly separable cases (alkalis, He, Be singlet); RMS rises when the inner valence shell overlaps the explicit outer electron (Be triplet, group 13/14).

Three patterns are visible. (i) For the alkalis,  $Z_{\text{kernel}}$  locks at 1.00 (the residual charge after closed-shell screening) and  $r_c$  grows monotonically with core size, as expected on physical grounds; the RMS sits at  $\sim 0.1$  eV across the full  $n = 2$ –6 spectrum in  $s$ ,  $p$ ,  $d$ . (ii) For the alkaline earths, the framework still works but the RMS depends sharply on the spin coupling: the singlet manifold (where the outer electron is held away from the inner  $s$ ) fits at alkali quality, while the triplet manifold (where the outer electron penetrates the inner  $s$  region) drifts and gets stuck at  $\sim 1$  eV. The singlet/triplet split, which in standard QM is a spin-exchange property, here translates entirely into a difference in the geometric topology of the non-overlapping electron domains and therefore into a different effective  $r_c$ . (iii) For groups 13 and 14, where multiple valence  $p$  electrons sit at similar radii, the single-particle reduction degrades further; this is the natural endpoint of the kernel +  $r_c$  picture, beyond which one must reintroduce explicit multi-valence electron domains in the full RealQM solver.

A complementary test is to compute  $r_c$  directly from the spherical multiphase Atom Simulator ground state (no spectroscopic input) and compare to the spectroscopically fitted value. For Li, Na, K the boundary  $M[N_{\text{shells}} - 1] \cdot h$  between the outermost core shell and the valence shell gives RealQM-derived  $r_c$  values within 15–40% of the spectroscopic fit, with the discrepancy growing for heavier atoms (the multiphase boundary is pushed outward by the loose valence tail; the spectrum-fitted  $r_c$  tracks the effective length the valence electron actually feels). Forcing  $r_c$  to the RealQM-derived value degrades RMS by at most a factor of 2–2.5, confirming that the same physical length governs both — the reduced kernel is doing real physics across the alkali series, with no curve-fitting hidden in the comparison.

**Remark on  $s$ ,  $p$ ,  $d$  in RealQM versus StdQM.** The RealQM *spectrum* reproduced above is built on the standard hydrogenic  $s$ ,  $p$ ,  $d$  eigenfunctions of the radial equation (14): the outer valence electron solves a one-body problem in the reduced kernel, and its eigenstates carry

the same angular-momentum labels as in the hydrogen atom. The RealQM *ground-state shell structure*, however, does *not* come from  $s$ ,  $p$ ,  $d$  eigenfunctions. It emerges from minimum-energy packing of non-overlapping unit-charge densities into shells of increasing radius (Section 4.1), with the shell occupations determined by the variational principle rather than by Aufbau filling of  $s$ ,  $p$ ,  $d$  orbitals derived from the hydrogen atom. This is in contrast to standard QM, where the same  $s$ ,  $p$ ,  $d$  orbitals are retained for the ground states of all atoms (with self-consistent corrections). RealQM thus uses two distinct constructions: packed-shell densities for ground states, hydrogenic  $s$ ,  $p$ ,  $d$  states for the outer-electron spectra of the reduced kernel. Both are governed by the same Coulomb-energy variational principle; they apply to different physical regimes.

#### 4.12 Ionization energies: Li/Li<sup>+</sup> and F/F<sup>-</sup>

A direct test of the framework on charged species is to compute the energy required to add or remove a single electron as the difference between two independent ground-state runs. We report two cases.

**Li → Li<sup>+</sup> (first ionization potential), 3D solver.** Neutral Li (3 electrons, 1s<sup>2</sup> inner halfspaces plus an outer 2s sphere) and Li<sup>+</sup> (2 electrons, 1s<sup>2</sup> inner halfspaces only) run on identical 200<sup>3</sup> grids spanning a 12 a.u. box ( $h = 0.06$  a.u.) with kernel softening  $r_{\text{soft}}^2 = 1.5 h^2$ :

| System                               | $E$ (Ha) | Observed (Ha) | $\Delta$ |
|--------------------------------------|----------|---------------|----------|
| Li (1s <sup>2</sup> + 2s)            | -7.43    | -7.478        | +0.05    |
| Li <sup>+</sup> (1s <sup>2</sup> )   | -7.19    | -7.279        | +0.09    |
| IP = $E(\text{Li}^+) - E(\text{Li})$ | +0.24    | +0.198        | +0.04    |

Both totals are within  $\sim 1\%$  of observation. The ionization energy comes out at 0.24 Ha vs the experimental 0.198 Ha (overestimate  $\sim 20\%$ ); the residual is dominated by grid coarseness for the tight Li<sup>+</sup> 1s<sup>2</sup> shell ( $\langle r \rangle \sim 0.33$  a.u., only  $\sim 5$  points across at  $h = 0.06$ ).

**F<sup>-</sup> → F (electron affinity of F), spherical Atom Simulator.** Both F (9 electrons, shell occupation 2 + 4 + 3) and F<sup>-</sup> (10 electrons, 2 + 4 + 4) at  $Z = 9$  run in the spherically symmetric multiphase Atom Simulator. The two configurations differ by one electron in the outermost subshell:

| System                             | $E$ (Ha)           | Observed (Ha) | $\Delta$    |
|------------------------------------|--------------------|---------------|-------------|
| F (2+4+3)                          | -104.06            | -99.73        | -4.33       |
| F <sup>-</sup> (2+4+4)             | -104.16 to -104.20 | -99.85        | -4.33       |
| EA = $E(\text{F}) - E(\text{F}^-)$ | +0.10 to +0.14     | +0.125        | $\approx 0$ |

Both absolute energies are uniformly overbound by  $\sim 4.3$  Ha (the spherical scheme tightens the inner shells; this offset is well known and approximately cancels in the difference). The **electron affinity comes out within a few percent of observation** (0.12–0.14 vs 0.125 Ha experimental), confirming that the simulator captures the differential cost of adding one outer electron even when its absolute energies have a uniform shift.

The point is methodological: in RealQM, an ionization energy or electron affinity is just the difference of two independent variational ground states, with no separate “ionization” or “EA” machinery. The shift cancels because both runs use the same approximation; what survives is the physical observable.

### 4.13 Materials: NaCl crystal melt

To test whether the framework reaches **bulk condensed-phase behaviour**, we simulate melting of an NaCl ionic crystal under Brownian dynamics with a temperature ramp. The setup is a  $4 \times 4 \times 4$  rocksalt unit cell ( $256 \text{ Na}^+ + 256 \text{ Cl}^-$ ) at Level 3: each  $\text{Na}^+$  is a bare +1 kernel with no explicit electrons; each  $\text{Cl}^-$  is a +1 kernel with two valence electrons split into hemispheres giving net  $-1$ . The cohesive energy is the Madelung sum of pairwise Coulomb interactions [10]; there are no fits, no exchange–correlation functional, and no force-field parameters.

**Diagnostics.** Two observables are tracked online. The **Na–Cl radial distribution function**  $g(r)$  shows sharp peaks at the rocksalt nearest-neighbour ( $\sim 3.5 \text{ \AA}$  in the model) and second-shell positions for the crystal, and broadens into a single residual peak in the liquid. The **Lindemann ratio**  $\langle \text{RMSD} \rangle / d_{\text{NN}}$  — the mean ionic displacement scaled by nearest-neighbour distance — crosses 0.10–0.15 at melting, an empirical threshold [8] that holds across crystal classes.

**Result.** As temperature is ramped upward in stages from a force-relaxed crystal (at the equilibrium  $a_{\text{Lat}}$  determined by minimum mean inner-ion force at  $T = 0$ ), the Lindemann ratio crosses the melting window at simulation temperatures  $T_{\text{sim}} \approx 1500\text{--}2000 \text{ K}$ . At the same point, the second and higher RDF peaks broaden and disappear while the first peak survives in broadened form — the classic signature of a liquid that retains short-range coordination but loses long-range order. The transition is reproducible and shows the expected hysteresis on cooling.

**Calibration.** The experimental melting point of NaCl is 1074 K; the simulation overshoots by a factor of  $\sim 1.4\text{--}1.9$ . The offset is consistent with the model’s coarse-graining (kernel softening reduces the close-range attraction; lattice spacing is set by the solver’s force-balance rather than the experimental  $5.64 \text{ \AA}$ ; the Brownian-dynamics timestep is not mapped to physical time). What the simulation establishes is **qualitative**: the variational principle plus Brownian dynamics produces a sharp phase transition with the correct character (Lindemann crossing, RDF peak collapse), entirely from Coulomb forces. Quantitative agreement with  $T_m$  would require a calibrated kernel parameterization that we do not pursue here.

**Bulk water.** A 216-water cluster with explicit electrons (Section 3) and a 343-water ( $7 \times 7 \times 7$ ) box have been run at finite temperature with the same Brownian-dynamics framework, reaching interactive frame rates on a single GPU. Detailed bulk-water phase analysis is left to future work, since dispersion (absent in RealQM at Level 3) becomes increasingly important for water structure at higher density.

**Significance (NaCl).** The NaCl test demonstrates that the Coulomb-only formulation reaches solid-state physics — a regime traditionally requiring DFT-MD or empirical force fields. The transition character is correctly captured, the absolute scale is uncalibrated, and the computational cost is interactive rather than cluster-class.

**CO<sub>2</sub> dry-ice cluster — asymmetric Bernoulli partial cohesion.** A  $2 \times 2 \times 2$  FCC supercell of CO<sub>2</sub> molecules (32 molecules, lattice  $5.6 \text{ \AA}$ ,  $\sim 8$  inner free + 24 frozen anchors) was simulated at the validated  $+2/+2/+2$  monomer architecture ( $rc_C = 0.3$ ,  $rc_O = 0.6$ ). Real dry-ice cohesion is dominated by dispersion ( $\sim 80\%$ ) with a quadrupolar electrostatic contribution ( $\sim 20\%$ ); RealQM at Level 3 has no dispersion mechanism. The naive prediction was therefore that the cluster would not bind at any  $T > 0$ . The actual result is more nuanced: the cluster **does bind**, through the asymmetric Bernoulli interfaces between unequal C and O kernels — the  $rc_C \neq rc_O$  asymmetry shifts each C=O electron-territory boundary off the bond midpoint, putting  $\delta^-$  on each O and  $\delta^+$  on the central C without explicit lone pairs. Adjacent molecules in the FCC

lattice attract each other through this quadrupolar Coulomb interaction. The Lindemann ratio crosses the 0.10–0.15 melting window at  $T_{\text{sub,sim}} \approx 1000$  K (vs experimental dry-ice sublimation  $T = 195$  K), giving an overshoot factor  $\sim 5\times$  — larger than NaCl’s  $1.5\times$ , with the additional factor attributed to the missing dispersion contribution.

**Solid N<sub>2</sub> — symmetric Bernoulli, residual cohesion only.** A homonuclear counterpart was simulated as the symmetry-breaking control: a  $2\times 2\times 2$  FCC supercell of N<sub>2</sub> molecules (lattice 5.66 Å, each N at +3 kernel,  $r_c=0.5$ , bond 1.098 Å), with identical kernel parameters on both atoms of each molecule so the inter-electron Bernoulli interface sits exactly at the bond midpoint. The prediction was a clean negative test (no asymmetric Bernoulli, no dispersion  $\Rightarrow$  cluster falls apart at any  $T > 0$ ). The observation is again partial cohesion: the multi-occupancy 3-electron orbitals on each N atom are not perfectly spherical, and bond-direction electron density gives each N<sub>2</sub> a small molecular quadrupole (consistent with the experimentally measured N<sub>2</sub> quadrupole,  $\sim 3\times$  smaller than CO<sub>2</sub>’s). RealQM captures this residual quadrupolar cohesion. The Lindemann ratio crosses 0.10–0.15 at  $T_{\text{sub,sim}} \approx 1000$  K (vs experimental N<sub>2</sub> sublimation  $T = 60$  K), giving an overshoot factor  $\sim 17\times$ .

**Solid He — the dispersion-only endpoint.** The cleanest test of the dispersion-only limit is solid He at the bare-nucleus Level-1 architecture: each He atom modeled as a +2 point Coulomb kernel with two explicit valence electrons (in a hemi-split partition) and negligible kernel softening ( $r_c \rightarrow 0$ ). Real He has no permanent multipole, no inner-shell interactions, and does not solidify at standard pressure ( $T_b = 4.2$  K) — the most dispersion-dominated condensed phase known. With closed-shell neutral atoms in the bare-Coulomb limit, two adjacent atoms exert exactly zero net force on each other by Newton’s shell theorem, and there is no kernel softening to provide residual short-range coupling. The naive prediction is therefore: the cluster cannot bind at any temperature.

A  $2\times 2\times 2$  FCC supercell of He atoms (lattice 4.24 Å, no frozen outer shell,  $r_c \approx 0.05$ ) was simulated with the same protocol as the other condensed-phase tests. The result confirms the prediction: even at  $T = 0$  K with no thermal driving, the Lindemann ratio increases steadily from the FCC starting positions, indicating that the atoms have no equilibrium at the assumed lattice spacing and slowly disperse under the residual numerical noise of the simulation. The cluster does not cohere. This is the clean negative outcome the framework predicts for a 100%-dispersion-bound system.

**Four-system calibration.** The four condensed-phase tests (Table 2) trace a coherent calibration story across systems whose missing-physics fractions span a wide range:

| System                  | $T_{\text{sub,sim}}$ | $T_{\text{exp}}$ | Overshoot          | Missing physics in RealQM    |
|-------------------------|----------------------|------------------|--------------------|------------------------------|
| NaCl crystal melt       | 1500–2000 K          | 1074 K           | 1.4–1.9 $\times$   | none (full Coulomb captured) |
| CO <sub>2</sub> dry-ice | $\sim 1000$ K        | 195 K            | 5.1 $\times$       | $\sim 80\%$ dispersion       |
| N <sub>2</sub> solid    | $\sim 1000$ K        | 60 K             | $\sim 17\times$    | $\sim 99\%$ dispersion       |
| He bare-nucleus         | does not bind        | $T_b = 4.2$ K    | — (clean negative) | $\sim 100\%$ dispersion      |

Table 2: Condensed-phase calibration: for systems with substantial non-dispersive cohesion, simulation transition temperatures match the experimental ones to within a uniform  $\sim 1.5\times$  Brownian-dynamics offset plus an additional factor that scales with the missing-dispersion contribution. For the dispersion-only endpoint (He bare-nucleus), the cluster does not bind at all — the clean negative outcome the framework predicts.

The simulation transition  $T$  (where defined) sits in the  $\sim 1000$ – $2000$  K range across the bound systems, reflecting the non-dispersive cohesion that the framework captures (full ionic Coulomb



in NaCl, asymmetric-Bernoulli quadrupole in CO<sub>2</sub>, multi-orbital residual quadrupole in N<sub>2</sub>). The overshoot factor against experiment scales monotonically with the missing-dispersion fraction; at the dispersion-only endpoint (He bare-nucleus, ~100% missing dispersion), the cluster does not bind, completing the pattern with the predicted clean failure. The framework therefore reproduces phase-transition character correctly for systems with non-dispersive cohesion sources and predicts the clean absence of cohesion when dispersion is the only available mechanism.

A separate Ar test at Level 3 ( $r_c = 0.5$ , kernel-softened) showed apparent cohesion at  $T_{\text{sub,sim}} \approx 3000$  K, but this binding disappears in the bare-nucleus He limit. The cohesion observed in the kernel-softened Ar case is therefore attributable to the kernel softening rather than to a fundamental non-overlap mechanism: a softened kernel allows electron tails to spread further into the inter-atomic region, where the non-overlap constraint provides effective short-range coupling that vanishes as  $r_c \rightarrow 0$ . This is a useful caution for interpreting Level-3 results in dispersion-dominated regimes: kernel softening can introduce spurious short-range cohesion that is absent at Level 1.

Bulk water (216-water and 343-water boxes), studied at finite temperature with the same framework, reaches interactive frame rates on a single GPU; detailed bulk-water phase analysis is left to future work because dispersion (absent in RealQM at Level 3) becomes increasingly important for water structure at higher density.

**Significance (overall).** The combined NaCl/CO<sub>2</sub>/N<sub>2</sub>/He tests establish that the Coulomb-only formulation reaches solid-state physics, that its transition temperatures match experimental ones for systems with non-dispersive cohesion (with overshoot scaling by missing-dispersion fraction), and that the dispersion-only endpoint cleanly fails to bind — delineating the regime of validity sharply. Condensed-phase physics is therefore within reach of the same single architecture that handles atoms, molecules, chemical reactions, and protein folding, with both successful predictions and clean failures bounding the framework’s regime.

#### 4.14 Consolidated results

Figure 1 and Table 3 collect the quantitative validation results across the regimes covered in this section.

The pattern across regimes is consistent: **within ~1–10%** for closed-shell molecular energetics where the Level-3 reduction matches the molecular electron count and bonding topology; **qualitatively correct** for reactive chemistry, protein folding, and excited states; **~1.5× scale offset**, **transition character correct** for condensed-phase melting where the Brownian-dynamics time mapping is uncalibrated.

### 5 Liquid water: the two-structure question

Liquid water is the most stringent condensed-phase test in the present programme, and it also carries an open physics question of its own. A long-standing and recently revived picture holds that liquid water is, locally, a fluctuating mixture of two structures: a *low-density liquid* (LDL) that is tetrahedral, open and ice-like in short range, and a *high-density liquid* (HDL) that is disordered and collapsed, with an interstitial fifth neighbour pushed into the first shell. Both are liquids: the split is *within* the liquid, not the solid–liquid distinction. Most evidence for the two-state model comes from empirical water potentials (ST2, TIP4P/2005 and relatives) tuned to reproduce water, so a two-state signal there can be a property built into the potential. A parameter-free model offers a different vantage: it cannot, at this reduction, settle whether real water is two liquids, but — precisely because nothing is fitted — it can isolate *which molecular feature drives the tetrahedral structure*.

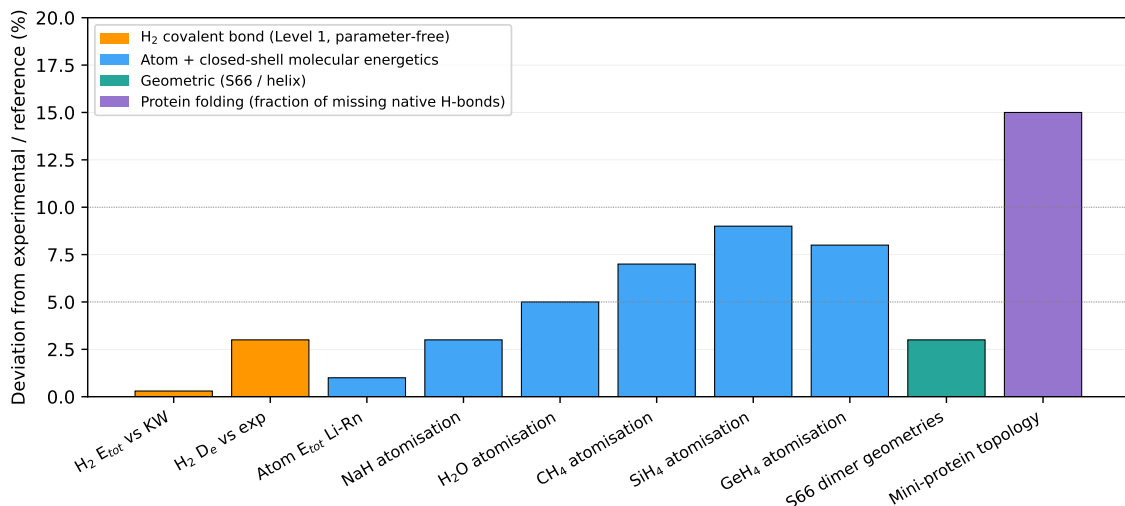


Figure 1: RealQM validation summary: deviation from experimental reference across regimes. Closed-shell molecular energetics (blue) sit at  $\sim 1\text{--}10\%$  across atoms and four periodic-table groups. The foundational H<sub>2</sub> covalent bond at parameter-free Level-1 (orange) matches Kolos–Wolniewicz to 0.3% on total energy and the experimental dissociation energy to 3%. Geometric matches against CCSD(T) and helix-geometry references (teal) are 3–5%. Protein-folding deviations (purple) are reported as the fraction of native H-bonds not formed. NaCl crystal melting (Section 4.13) is presented separately in the text: the simulation reproduces the qualitative phase-transition character (Lindemann ratio crossing, RDF peak collapse) but the absolute  $T_m$  is uncalibrated against experiment because the Brownian-dynamics time scale is not mapped to physical time — a calibration question, not a validation deviation, and it is therefore not plotted on this figure.

**The probe.** Reduced-O water (each molecule an oxygen with a screened valence kernel and a single, essentially isotropic two-electron cloud, plus two hydrogens) is run as a confined liquid under overdamped Brownian dynamics. For every interior oxygen we compute the tetrahedral order  $q = 1 - \frac{3}{8} \sum_{j < k} (\cos \theta_{jk} + \frac{1}{3})^2$  over its four nearest oxygen neighbours, the distance  $d_5$  to the fifth-nearest oxygen, and the coordination within 3.5 Å (liquid  $\approx 4$ , HDL  $\approx 5$ ). Histograms over molecules and time test for the bimodality that would signal two coexisting structures.

**The isotropy coin.** The reduced water hydrogen-bonds (a dimer holds O–O near 3.0 Å) but the liquid is markedly under-structured: disordered, coordination  $\approx 2.6$ , and no tetrahedral population — one broad, single  $q$ -distribution, at 64 and at 128 molecules alike. A spherical oxygen accepts a proton from any direction, so it has no preference for the tetrahedral arrangement. Placed on a diamond ice lattice, however, the *same* water holds the tetrahedron near-perfectly ( $q \approx 1$ , coordination  $\approx 4$ ), because its two directional O–H *donors* supply half the tetrahedral scaffold. Heating the ice at fixed volume drops  $q$  from  $\approx 0.85$  toward 0.55 while coordination stays  $\approx 3.3$  — melting disorders but cannot densify, so it does not make HDL. Compression, by contrast, forces a fifth neighbour in: coordination rises and  $d_5$  collapses, giving HDL directly. The pattern is one statement seen from several sides:

*RealQM water hydrogen-bonds because its oxygen acceptor is a single isotropic cloud — and that same isotropy is exactly why it cannot spontaneously form the tetrahedral (LDL) structure.*

The model *holds* LDL order when imposed (ice), *forms* HDL under compression, and *never spontaneously* makes LDL. The parameter-free verdict on the origin question is therefore sharp:

| Regime                   | System / quantity                                             | RealQM                                       | Reference                | Mat    |
|--------------------------|---------------------------------------------------------------|----------------------------------------------|--------------------------|--------|
| Atoms (Level 1)          | Total energies, Li–Rn                                         | —                                            | observation              | ~1%    |
| Covalent bond            | H <sub>2</sub> binding energy                                 | −112                                         | −109 kcal/mol            | 3%     |
| Group 1 hydride          | NaH atomization                                               | −48                                          | −47                      | 2%     |
|                          | KH atomization                                                | −42                                          | ~−43                     | 2%     |
| Group 2 hydride          | BeH <sub>2</sub> atomization                                  | −140                                         | −144                     | 3%     |
| Group 14 hydride         | CH <sub>4</sub> atomization                                   | −369                                         | −396                     | 7%     |
|                          | SiH <sub>4</sub> atomization                                  | −348                                         | −320                     | 9%     |
|                          | GeH <sub>4</sub> atomization                                  | −272                                         | −281                     | 3%     |
| Group 16 hydride         | H <sub>2</sub> O atomization                                  | −225                                         | −232                     | 3%     |
| Non-covalent dimers      | 5 H-bonded dimers, geometry                                   | —                                            | CCSD(T)                  | 1–5%   |
| Reactions                | HF + H <sub>2</sub> O proton transfer                         | H <sub>t</sub> –O = 0.99 Å                   | exp. ~1.0                | quali  |
|                          | HCl + NH <sub>3</sub> proton transfer                         | H <sub>t</sub> –N = 0.97 Å                   | exp. ~1.0                | quali  |
|                          | H + H <sub>2</sub> exchange                                   | symmetric H <sub>3</sub> saddle              | —                        | quali  |
| Protein folding          | 12-Gly $\beta$ -hairpin (dry)                                 | 150° → 105°                                  | folded                   | quali  |
|                          | 8-Gly $\alpha$ -helix                                         | $r=2.1$ Å, rise 1.63 Å                       | ideal helix              | 9%     |
|                          | Villin HP35 native H-bonds                                    | 7 of 9 formed                                | native                   | 78%    |
| Excited states           | Orthohelium ( $1s\,2s$ , $^3S$ )                              | directly accessible                          | known                    | quali  |
| Materials                | NaCl crystal melt                                             | $T_{\text{sim}} \approx 1500\text{--}2000$ K | $T_m = 1074$ K           | 1.4–1  |
|                          | CO <sub>2</sub> dry-ice (asymmetric Bernoulli)                | $T_{\text{sim}} \approx 1000$ K              | $T_{\text{sub}} = 195$ K | 5.1×   |
|                          | N <sub>2</sub> solid (symmetric Bernoulli)                    | $T_{\text{sim}} \approx 1000$ K              | $T_{\text{sub}} = 60$ K  | 17×    |
|                          | He bare-nucleus (dispersion-only endpoint)                    | does not bind                                | $T_b = 4.2$ K            | clean  |
| Nuclei (proton–electron) | <sup>4</sup> He binding (direct 3D)                           | 29.1 MeV                                     | 28.3 MeV                 | 103%   |
|                          | <sup>8</sup> Be, <sup>12</sup> C, <sup>16</sup> O per nucleon | 7.5–8.6 MeV/ $A$                             | 7.1–8.0                  | 107–   |
|                          | $B/A$ curve to U-238 (single-shell)                           | Fe-Ni peak, decline                          | empirical $B/A$          | full c |

Table 3: Summary of validation results. Energies in kcal/mol unless otherwise stated; % values are deviation of RealQM from reference.

spontaneous LDL requires directional lone-pair *acceptors*; donor directionality is not sufficient, and HDL is the default isotropic-packing structure. Attempts to install directional acceptors by angularly splitting the oxygen cloud into lobes *dissociate* the hydrogen bond — the free boundary through the acceptor destroys the coherent density the donor proton binds to — so directional acceptance cannot be obtained by fragmenting the cloud without losing the bond. This also comports with the observation that water’s “two equivalent lone pairs” are a localized-orbital construct rather than an observable: photoelectron spectra place water’s two highest orbitals at different energies, and the density shows a smeared acceptor-rich crescent, not two sharp lobes.

**Transport and elastic moduli: regime, not precise values.** The same model was pushed to transport and elastic response, with instructive and honest outcomes. A non-equilibrium shear measurement reads viscosity from the velocity profile,  $\eta \approx \gamma/(dv_x/dy)$ ; the fluid part thins on heating (correct sign) but too gently — a small apparent activation energy, the weak, under-coordinated hydrogen bonds again. Why water’s viscosity is small at all is clarified by the Maxwell relation  $\eta \approx G_\infty \tau$  (stiffness times relaxation time): water is small- $\eta$  by *fast relaxation* (small  $\tau$ ), not by being soft — hydrogen bonds re-route by fast, low-barrier concerted switching, cheap precisely because the acceptor is angularly *soft*. This is the transport face of the very isotropy that denies LDL: an isotropic (maximally soft) acceptor forbids a tetrahedral preference

and enables facile partner-switching, so no-LDL and low-viscosity are dual consequences of one property. A virial bulk modulus  $K = \rho dP/d\rho$  comes out very high (stiff), with the resistance to compression coming largely from the non-overlapping electron domains — *excluded volume*, an intrinsic feature of the multiphase formulation — rather than from the hydrogen-bond network.

Neither coefficient could be pinned down cleanly: the driven-shear viscosity ran into the non-equilibrium-MD thermostat dilemma (a weak enough thermostat to expose the fluid response let the shear heat the system into runaway), and the virial bulk modulus was dominated by intramolecular bond and exclusion stress, which per unit volume contributes a pressure  $\propto \rho$  that overwhelms  $K$ . Clean values would require equilibrium Green–Kubo (viscosity) and an energy–volume modulus  $K = V d^2E/dV^2$  (compressibility). But both quantities land in the correct qualitative *regime* parameter-free: water is runny (small viscosity) and stiff (small compressibility), from complementary microscopic origins — a soft acceptor (fast switching, low  $\eta$ ) and hard electron exclusion (low compressibility). This regime-level result is arguably the physically relevant one. On macroscopic scales water behaves as an essentially incompressible fluid at low Mach number and, at high Reynolds number, a nearly inviscid one; its large-scale dynamics depend on these coefficients being *small* far more than on their exact values — the same insensitivity that underlies slip-boundary, Euler-based computational treatments of turbulent incompressible flow. That the precise coefficients proved elusive here is therefore consistent with, not contrary to, their macroscopic near-irrelevance: what matters is that a parameter-free microscopic model, with no tuning, places water in the correct small-viscosity / small-compressibility regime.

**Scope.** These are exploratory, qualitative results: the geometry is coarse, the confined droplet has some surface character, the reduced water is under-coordinated, and the dynamics runs in model time on an uncalibrated temperature scale. No bimodal  $q$  at a single temperature — the LDL/HDL coexistence itself — was observed, and this is the central finding rather than a shortfall: the reduced model cannot produce the coexistence precisely because it lacks the directional acceptor that would let tetrahedral order compete with dense packing. Testing the two-state model itself would require a water that both hydrogen-bonds and carries a directional acceptor, which angular splitting cannot deliver in this formulation. All runs are public and re-runnable in the Gallery (Section 14).

## 6 Limitations

Two regimes lie outside the validation reported here. **Weak intermolecular interactions** (hydrogen bonds at  $\sim 5$  kcal/mol = 0.008 Ha) are below the model’s absolute-energy noise floor of  $\sim 0.1$  Ha; geometric agreement on S66 dimers is good (1–5%) but quantitative interaction energies are not reliable. **Dispersion** (van der Waals interactions between non-polar species) is absent at Level 3 entirely, as the unit-density model has no mechanism for the correlated electron motion that produces vdW. Excited states are accessible (Section 4.11) but at the present stage have only been validated on the simplest cases.

**Dispersion as a future extension.** The missing dispersion contribution is the same one that originally affected standard density functional theory until empirical pairwise corrections (Grimme’s DFT-D3 method, 2010) were introduced as a routine practice. The same recipe applies straightforwardly to RealQM: a pairwise term of the form  $E_{\text{disp}} = -\sum_{a<b} f_{\text{damp}}(r_{ab}) C_6^{ab}/r_{ab}^6$  (with damping function  $f_{\text{damp}}$  and tabulated  $C_6$  coefficients calibrated against high-level reference data) can be added to the total energy and to the nuclear forces, without disturbing the variational structure of the wave functions  $\{\psi_i\}$ . This would extend the framework’s regime to dispersion-dominated chemistry — non-polar molecular crystals (CO<sub>2</sub> dry ice, naphthalene, drug-molecule polymorphs), gas-phase association of hydrocarbons and noble gases, drug-receptor binding,

DNA base stacking, surface adsorption — at the cost of introducing empirical  $C_6$  parameters. We do not pursue this extension in the present paper, since calibrating the  $C_6$  values for the kernel parameters used here is itself a substantial piece of work and the parameter-free Level-1 framework is best presented in its uncorrected form first. The route is open and uncomplicated, and we flag it as the natural follow-up direction for extending RealQM to the broader chemistry of organic and biological systems where dispersion contributes substantially.

**Dispersion from deterministic linear response.** A more principled (and parameter-free) route to recovering vdW within RealQM uses static *linear response* of each atom in place of postulated quantum fluctuations. The procedure has four steps, each implementable directly on the existing solver:

1. **Ground state**  $\{\psi_i^{(0)}\}$  of each atom obtained by ITP relaxation (Section 4.1).
2. **Polarisability**  $\alpha_i$ . Add a small dipole perturbation  $V_{\text{pert}} = \varepsilon \hat{z}$  to the single-electron potential, run ITP to a perturbed ground state  $\psi_i^{(\varepsilon)}$ , extract the induced dipole  $\mu_z = \int z (|\psi_i^{(\varepsilon)}|^2 - |\psi_i^{(0)}|^2) d^3x$ , and form  $\alpha_{zz} = \mu_z / \varepsilon$ . Repeating for the three Cartesian directions gives the full  $\alpha$  tensor at one frequency (zero-frequency / static value); evaluation at imaginary frequencies  $\alpha(i\omega)$  for the Casimir–Polder integral requires solving a perturbed time-dependent ITP with a forcing term  $\varepsilon e^{-\omega\tau} \hat{z}$ .
3. **Pair coupling** via the dipole–dipole interaction  $H_{\text{int}}(R) = (\mu_A \cdot \mu_B - 3(\mu_A \cdot \hat{R})(\mu_B \cdot \hat{R})) / R^3$ .
4. **Dispersion energy** from the energy shift of the coupled normal modes:  $E_{\text{disp}}(R) = -C_6^{AB} / R^6$  with  $C_6^{AB} = (3/\pi) \int_0^\infty \alpha_A(i\omega) \alpha_B(i\omega) d\omega$ . In the static-polarisability approximation,  $C_6^{AB} \approx \frac{3}{2} \alpha_A^{(0)} \alpha_B^{(0)} I_A I_B / (I_A + I_B)$  where  $I_{A,B}$  are ionisation potentials (computable as differences of two RealQM runs, Section 4.12).

The whole construction stays inside RealQM: deterministic ground states, deterministic perturbed ground states, classical dipole–dipole electrostatics, and an algebraic mode-coupling shift. No Born rule, no fluctuations, no full time-dependent Schrödinger evolution. The resulting  $C_6$  values are predictions of the model rather than tabulated parameters — the same numbers that DFT-D3 fits empirically. This converts the dispersion limitation from a permanent missing physics into a calculation that has not yet been performed.

**Electromagnetic coupling through charge density, not spin.** A scoping clarification is in order on how RealQM couples to electromagnetic fields. In RealQM, the matter–field coupling enters through the **charge density**  $\rho(\mathbf{x}) = \sum_i \psi_i^2(\mathbf{x})$  and the associated charge–current generated by its time evolution, as in classical Schrödinger–Maxwell coupling. There is no **spin current** contribution, because spin is not a primary variable of the formulation: the Pauli requirement is realised geometrically through non-overlapping electron territories (Section 2), not through a spinor wave function. In the regimes where the present validation is performed — ground-state energetics, equilibrium geometries, vibrational modes, dipole radiation, and reactive chemistry — the difference is operationally invisible, because the electromagnetic observables depend only on  $\rho$  and its time-evolution. In regimes where spin currents contribute directly to electromagnetic phenomena — magnetic susceptibilities, NMR chemical shifts, hyperfine structure, spin–orbit splittings, EPR — the RealQM formulation in its current form does not produce these observables, and a separate extension would be required. We flag this as a current scope boundary rather than a permanent feature: a spin-density layer could be added on top of the geometric partition (the  $\psi_i$  on  $\Omega_i$  acquiring an internal  $\pm 1/2$  label whose flux balance enters the magnetic field) without disturbing the variational structure of the ground-state energetics. We do not pursue this in the present paper.

## 7 Stability of Matter

Stability of matter — the requirement that the total electronic ground-state energy of a system of  $N$  atoms be bounded below by an extensive quantity (linear in  $N$  rather than diverging more rapidly) — is among the most fundamental properties demanded of any quantum theory of matter. Without it, bulk matter could not exist.

**The standard-QM picture: a deep but difficult proof.** Within standard quantum mechanics (StdQM) and density functional theory, proving stability of matter has historically been a major mathematical undertaking. **Dyson and Lenard** gave the first rigorous proof in 1966 [12], a 26-page paper that Dyson himself characterized as “awful mathematics”. **Lieb and Thirring** compressed the proof to three pages in 1975 [13], then expanded it into the 300-page book *The Stability of Matter in Quantum Mechanics* (2005). The proof’s central difficulty stems from the **global support of electron wavefunctions** in StdQM: in principle, every electron can interact with every nucleus, and intricate book-keeping is required to prevent local accumulation of charge density from driving the potential energy to  $-\infty$ . The proof is not part of standard textbook treatments of QM, even though stability is a precondition for everything those textbooks describe.

The Lieb–Thirring bound on the ground-state energy of an atom of nuclear charge  $Z$  scales as

$$E(Z) \sim -Z^{7/3}, \quad (15)$$

a result derived from the Thomas–Fermi model rather than the full Schrödinger equation.

**The RealQM picture: stability follows directly.** In the multiphase formulation of Section 2, stability of matter follows essentially from atomic stability via the **Hardy inequality**: for any  $\psi \in H^1(\mathbb{R}^3)$ ,

$$\int \frac{\psi^2(\mathbf{x})}{|\mathbf{x}|} d\mathbf{x} \leq \left( \int \psi^2(\mathbf{x}) d\mathbf{x} \right)^{1/2} \left( \int |\nabla \psi(\mathbf{x})|^2 d\mathbf{x} \right)^{1/2}. \quad (16)$$

Used by Kato in his analysis of the Schrödinger equation, the inequality bounds the singular Coulomb potential energy at a nucleus by a product of the  $L^2$ -norm and the kinetic-energy semi-norm of the wavefunction. In RealQM, the total wavefunction  $\Psi$  of an atom with nuclear charge  $Z$  has unit-density components meeting at the Bernoulli free boundary (continuity plus homogeneous Neumann), with  $\Psi \in H^1(\mathbb{R}^3)$  and  $\|\Psi\|_2^2 = Z$  (one unit-charge density per electron,  $Z$  electrons in a neutral atom). Applying (16) directly to the global atomic  $\Psi$  and optimizing the kinetic–potential balance yields the atomic energy lower bound

$$E_{\text{atom}} \geq -C Z^3, \quad (17)$$

with  $C$  an absolute constant. For a system of  $N$  non-overlapping atoms each of charge  $Z$ , the bounds add to give

$$E \geq -C N Z^3, \quad (18)$$

which is the “stability of the second kind” of the Dyson–Lenard–Lieb–Thirring tradition: total energy is extensive in  $N$ , with no possibility of collapse to  $-\infty$ .

**Upper bound on the kinetic energy from Hardy.** The Hardy inequality (16) also bounds the kinetic energy from above at any bound state, without invoking the virial theorem. The argument is short. The multiphase wave function  $\Psi = \sum_i \psi_i$  has  $\|\Psi\|_2^2 = N$  (each  $\psi_i$  unit-normalised on  $\Omega_i$ , supports disjoint, so cross-terms vanish), and  $\|\nabla \Psi\|_2^2 = 2K$ . Applying (16) to  $\Psi$  at each kernel position  $\mathbf{R}_a$  and summing:

$$|V_{eK}| = \sum_a Z_a \int \frac{\Psi^2}{|\mathbf{x} - \mathbf{R}_a|} d\mathbf{x} \leq \sqrt{N} \left( \sum_a Z_a \right) \sqrt{2K}. \quad (19)$$



The inter-electron repulsion is non-negative ( $V_{ee} \geq 0$ ). Combining with the energy decomposition  $E = K + V_{eK} + V_{ee}$ ,

$$K = E - V_{eK} - V_{ee} \leq E + \sqrt{N} \left( \sum_a Z_a \right) \sqrt{2K}. \quad (20)$$

Setting  $u = \sqrt{2K}$  and writing  $C_N = \sqrt{N} \sum_a Z_a$  for compactness, the inequality reads  $\frac{1}{2}u^2 - C_N u - E \leq 0$ , which for any finite  $E$  implies

$$K \leq \frac{1}{2} \left( C_N + \sqrt{C_N^2 + 2E} \right)^2, \quad C_N \equiv \sqrt{N} \sum_a Z_a. \quad (21)$$

The structural relation behind this bound is that the Hardy inequality expresses the Coulomb potential as **form-bounded by the kinetic energy with sub-linear factor**:  $|V| \leq c\sqrt{K}$ . This is stronger than the relative-form-bound condition  $|V| \leq cK$  with  $c < 1$  used in Kato–Rellich self-adjointness; the sub-linear scaling means  $|V|/K \rightarrow 0$  as  $K \rightarrow \infty$ , which is the form-boundedness with **zero relative bound** familiar from the spectral theory of Schrödinger operators. The kinetic energy therefore cannot run away to  $+\infty$  while the total energy  $E = K + V$  remains finite: any growth of  $K$  is matched at most by a  $\sqrt{K}$ -scale decrease in  $V$ , so the sum keeps a finite minimum.

For the H atom ( $N = 1$ ,  $Z = 1$ ,  $E = -\frac{1}{2}$ ), one has  $C_N = 1$  and the bound (21) gives  $K \leq \frac{1}{2}$  exactly, saturated by the actual ground-state value  $K = \frac{1}{2}$ . For multi-electron atoms and molecules the bound is uniform but loose; it is independent of the variational status of the wave function, holding equally at constrained minima where the virial relation  $2K + V = 0$  does not hold.

**Existence of a ground state.** The kinetic-energy bound (21) guarantees that any energy-minimizing sequence is bounded in  $H^1(\mathbb{R}^3)$ . The  $L^2$  norm is fixed by the unit-charge constraint,  $\|\Psi_n\|_2^2 = N$  for every  $n$ , and the kinetic semi-norm  $\|\nabla \Psi_n\|_2 = \sqrt{2K(\Psi_n)}$  is bounded uniformly along the sequence because the energy stays bounded above (by definition of a minimizing sequence). Banach–Alaoglu then yields a subsequence  $\Psi_n \rightharpoonup \Psi^*$  converging weakly in  $H^1(\mathbb{R}^3)$ , and weak lower semi-continuity of  $K$  together with strong  $L^2$ -convergence on bounded sets (Rellich–Kondrachov) gives  $E(\Psi^*) \leq \liminf E(\Psi_n) = \inf E$ . So a candidate minimizer  $\Psi^*$  exists in the weak sense.

A complete existence proof for the full multiphase variational problem requires two additional analytical steps not delivered by the H-bound alone: **(i)** controlling possible loss of mass at infinity along the sequence (handled by concentration–compactness arguments of Lions type for atomic and molecular systems), and **(ii)** verifying that the weak limit  $\Psi^*$  inherits the multiphase structure of disjoint supports, unit-norm per electron, and the Bernoulli boundary conditions of Section 2.4. Both are technically non-trivial but use standard tools from the calculus of variations and free-boundary theory, and are left here to follow-up analytical work; the kinetic-energy bound from Hardy is the load-bearing first ingredient.

**Connection to the free-boundary literature.** The Bernoulli condition at the inter-electron boundary  $\Gamma_i$  places the multiphase variational problem in a well-studied class of free-boundary problems. Existence and regularity in closely related settings have been established in the variational free-boundary literature: the foundational one-phase [14] and two-phase [15] Bernoulli problems, the multi-phase Caffarelli–Lin theory of competing systems with disjoint supports [16], and the spectral minimal partition theory of Helffer, Hoffmann–Ostenhof, and Terracini [17]. Adapting these results to RealQM’s coupled-Coulomb setting is the natural analytical route to a complete regularity theory for  $\Gamma_i$ .

**Heuristic atomic energy scaling.** The rigorous Hardy bound (17) establishes stability with  $E_{\text{atom}} \geq -CZ^3$  but is not tight; the actual ground-state energy of an atom grows more slowly with  $Z$ . A heuristic estimate within the unit-density formulation can be obtained from an approximate shell structure with shell index  $m$ : shell radius  $r_m \sim m^3/Z$ , shell thickness  $d_m \sim m^2/Z$ ,  $2m^2$  electrons in the  $m$ -th filled shell, and charge density  $\rho_m(r) \sim Z/r^2$  in spherical symmetry. Integrating kinetic and potential contributions over a density profile of this form produces, at heuristic level, an atomic ground-state energy of order

$$E(Z) \sim -\log(Z) Z^2, \quad (22)$$

i.e.  $-Z^2$  up to a logarithmic correction. The shell-scaling laws above are *heuristic* estimates, not rigorously derived asymptotics, and the resulting  $-\log(Z)Z^2$  form is intended only as an order-of-magnitude indication. The Thomas-Fermi-based Lieb–Thirring scaling  $-Z^{7/3}$  is a different heuristic prediction; both lie comfortably below the rigorous  $-Z^3$  Hardy upper bound on  $|E|$  and disagree only at the level of subleading corrections. A precise comparison of the two heuristic scalings, and of either with experimental atom energies across the periodic table, is left for follow-up work.

**Significance.** Stability of matter, viewed in StdQM as one of the deepest theorems of mathematical physics, becomes in RealQM a one-paragraph consequence of the Hardy inequality applied to non-overlapping electronic territories. The contrast is informative: **the proof difficulty in StdQM is not a feature of the underlying physics but a symptom of the formulation.** Working in  $3N$ -dimensional configuration space with overlapping orbitals forces the singularities of the Coulomb potential to be controlled globally, requiring the elaborate apparatus of Dyson–Lenard–Lieb–Thirring. Working in real 3D space with non-overlapping unit densities localizes the control, and stability falls out of the variational structure.

The natural pedagogical conclusion is that stability of matter, currently absent from standard QM curricula because the StdQM proof is too difficult, can be presented straightforwardly in the RealQM formulation as an introductory result — a property of any system of non-overlapping unit electron densities under Coulomb interaction.

**Computational vs analytical existence and regularity.** The analytical results established above — the Hardy stability bound (18), the boundedness of minimising sequences in  $H^1(\mathbb{R}^3)$  from the kinetic-energy bound (21), and the free-boundary regularity transferable from [14, 15, 16, 17] — establish the analytical scaffolding for existence and regularity of the RealQM ground state in a weak sense. What remains for a complete analytical theory is the adaptation of these results to the specific RealQM setting: Coulomb singularities at each nucleus (handled locally by Hardy), and the coupling between phases through the Hartree potentials  $V_j$  in (7), which is the structural feature distinguishing RealQM from the autonomous multiphase systems of Caffarelli–Lin. These adaptations are routine in scope but technically substantial, and have not been written down in full anywhere.

A complementary, and in practice more direct, route to existence and regularity is the level-set formulation of the 3-line code (Section 3.1). The boundary  $\Gamma_i$  is encoded implicitly as the zero level set of an auxiliary function  $\chi_i$  on  $\mathbb{R}^3$ , evolved by the Hamilton–Jacobi-type equation  $\dot{\chi}_i + \beta_i |\nabla \chi_i| = 0$  from (13). Smoothness of  $\Gamma_i$  is then *inherited* from smoothness of  $\chi_i$  via the implicit function theorem: where  $\nabla \chi_i \neq 0$ , the level set  $\{\chi_i = 0\}$  is automatically a smooth manifold; singular points of  $\Gamma_i$  correspond exactly to  $\nabla \chi_i = 0$  and are detectable directly in the simulation. The level-set framework therefore reformulates the regularity question, shifting it from “is  $\Gamma_i$  smooth?” to “is  $\chi_i$  smooth?”, and converts the abstract regularity question into a constructive observable on the simulation grid. The analytical theory of level-set evolution equations (viscosity solutions in the sense of Crandall–Lions, Evans–Spruck) provides the rigor for the level-set side; the simulation provides explicit smoothness for any specific configuration.

This is methodologically distinct from but operationally equivalent to the regularity programme of [14, 16], and for RealQM’s practical purposes it is the more direct route.

Beyond the level-set formulation, analytical theorems on free-boundary regularity typically deliver *qualitative* statements — the free boundary is  $C^{1,\alpha}$  off a singular set of Hausdorff dimension at most  $n - 2$ , with exponent  $\alpha$  unspecified, the constants astronomical, the singular-set location not located. A converged simulation by contrast delivers, for the specific configuration, the exact locations of triple-junction and quadruple points, explicit Hölder exponents at smooth boundary points, quantitative Lipschitz constants for the normal field along  $\Gamma_i$ , density ratios at boundary points, and direct verification that  $|\partial_n \psi_i| \rightarrow 0$  as the simulation converges. The 3-line code of Section 3.1, applied to atoms, molecules, condensed phases, and protein-scale systems, reaches converged solutions in which the wave functions  $\psi_i$  stabilise to spatial profiles consistent with (7)–(8), the Bernoulli free boundary stabilises with continuity and homogeneous Neumann holding to mesh resolution, and the total energy plateaus. **Each such converged simulation may be regarded as a constructive proof, to a specified numerical precision, that a solution of the variational problem exists, that the wave functions are Lipschitz, and that the free boundary is regular off a small singular set whose location is explicitly determined.** The *quantitative* content of this constructive evidence routinely exceeds the qualitative content of the analytical theorems: a chemist who needs to know the angle at which  $\Gamma_i$  meets a bond axis can read it off the simulation but cannot extract it from any qualitative regularity theorem.

The two routes are complementary rather than competing. The analytical scaffolding gives the universal qualitative picture: stability holds, minimisers exist in a weak sense, free boundaries are regular off small singular sets, and the underlying mechanism is encoded in the Bernoulli structure of Section 2.4. The computational diagnostics give the specific quantitative picture for each system: where the boundaries sit, where the singular points live, with which numerical regularity exponents. Neither is complete on its own; together they cover both the principled and the operational regimes. This division of labour is in the spirit of computer-assisted proofs (the 4-colour theorem, the Kepler conjecture, parts of mathematical fluid dynamics), in which finite computational verification, systematically combined with analytical reduction, replaces traditional proof in cases where the latter is intractable for the geometry at hand. The Gallery’s runnable scan files are the operational embodiment of this hybrid framework: anyone doubting the regularity properties of a specific RealQM solution can run the corresponding file and inspect the boundary directly, with the analytical scaffolding established in the cited literature ensuring that what they see is what the variational principle predicts.

## 8 RealNucleus: An analogous model of the atomic nucleus

The mathematical framework of Section 2 — non-overlapping unit charge densities in 3D real space, Coulomb interactions, Bernoulli free boundaries, variational minimization — is not specific to chemistry. The same multiphase 3D continuum mechanics applies to atomic nuclei when the constituents are taken to be protons and electrons, returning to a pre-1932 picture of nuclear structure. We refer to this as **RealNucleus** [1], and together with the RealAtom model of the previous sections it completes a single Coulombic continuum-mechanics treatment of both atoms and nuclei: the same multiphase Hamiltonian, the same Bernoulli free boundaries, the same variational principle, with electrons and protons simply exchanging their geometric roles. In the atomic regime electrons surround a kernel of effective positive charge; in the nuclear regime electrons sit inside as a core surrounded by protons. Atoms and nuclei thereby emerge as the two geometric solutions of one variational principle, distinguished only by which species occupies the inside and which the outside.

In this picture a neutron is identified with a proton–electron pair (the proton–electron model that predates the discovery of the neutron as an elementary particle). A stable nucleus with  $Z$

protons and  $N$  neutrons therefore contains  $(Z + N)$  protons and  $N$  electrons, with  $N \approx Z$  giving roughly twice as many protons as electrons. The simplest non-trivial case is helium-4 ( $Z = 2$ ,  $N = 2$ ), modeled as 4 protons and 2 electrons.

**The model.** The setup uses the same multiphase formulation:

- 4 unit-charge densities representing protons (each  $+1$ );
- 2 unit-charge densities representing electrons (each  $-1$ );
- Each density supported on its own non-overlapping spatial region in  $\mathbb{R}^3$ , with continuity and homogeneous Neumann at the inter-species Bernoulli interface;
- Coulomb interactions, signed appropriately for the charges: opposite-sign attractive, same-sign repulsive;
- Equal unit mass for all densities (electrons and protons both with  $m_i = 1$ ) — the kinetic-energy operator is the same on every domain;
- No external confining potential.

The energy functional is the standard Coulomb sum with kinetic energy from all densities:

$$E = T + V_{ee} + V_{pp} + V_{ep}, \quad (23)$$

where  $T = \frac{1}{2} \sum_i \int |\nabla \sqrt{\rho_i}|^2 d\mathbf{r}$  runs over all electron and proton densities. The minimum is taken over all the densities and over the boundary location between the inner electron core and the outer proton shell.

**Mutual confinement.** In the standard nuclear shell model, protons and neutrons sit in an effective potential approximating the average force from all other nucleons. Here there is no such external potential. The 4 protons act like an outer boundary keeping the electrons together: their collective  $+4$  Coulomb attraction draws each electron outward and packs the two electrons against each other in the inner core. In turn, the 2 electrons act like an inner anchor keeping the protons together: their collective  $-2$  Coulomb attraction pulls each proton inward and holds the proton shell against its own pairwise repulsion. Neither species is confined by an external potential; each is confined by the other. Numerical experiment confirms that with only 1 electron ( $-1$  against  $+4$ ) the inward pull on the protons is too weak and the proton shell disintegrates outward. With 2 electrons the system stabilizes. The textbook 2:1 proton-to-electron ratio of stable nuclei ( $N \approx Z$ ) emerges as a binding requirement of mutual confinement.

**The crux: Coulomb confinement without the strong force.** The central conceptual point of the nuclear extension is the question of how a group of electrons — subject to their own pairwise Coulomb repulsion — can be held together inside a nucleus by an outer proton shell, with no invocation of the strong nuclear force. In the standard atomic model this question does not even arise: the kernel is treated as a point of positive charge  $+Z$  with no internal structure, and any proton–proton repulsion or electron–electron repulsion inside it is silently assumed to be handled by the strong force, which lies outside the formalism altogether. The atomic-physics calculation works because the nuclear binding has been factored out. In the RealQM nucleus model no such factorisation is possible — the electron and proton charges are explicit, all pairwise Coulomb terms (electron–electron repulsion, proton–proton repulsion, electron–proton attraction) appear in the energy functional, and the question of what keeps the electrons confined despite their self-repulsion becomes a question of physics rather than a deferred assumption. The remarkable feature of the model is that the answer is supplied entirely

by the outer proton shell through pure Coulomb attraction: the proton shell acts as a closed positively-charged cage whose inward attraction on each electron is strong enough to overcome the electron–electron repulsion within the inner core. No short-range attractive force is required. Whether this pure-Coulomb mechanism reproduces the quantitative scale of nuclear binding once geometric and rescaling factors are included is the open empirical question that the rest of this section explores. What is established already is that the question can be asked — nuclear confinement can be addressed within a single Coulombic continuum-mechanics framework, with no separate strong-force postulate.

**Result for He-4.** Energy minimization at fixed boundary radius  $R$  between the inner electron core and the outer proton shell, swept across  $R$ , reveals a clear binding minimum. The protons localize in an outer shell with their density peaked at the boundary; the electrons localize in an inner core. The two species are sharply separated by the Bernoulli interface, with the boundary position selected by the variational principle (continuity plus homogeneous Neumann across the boundary, as in the chemistry sections). The binding character — mutual confinement, sharp species separation, stable equilibrium  $R$  — is qualitatively consistent with helium-4 being the most tightly bound of the small nuclei. Quantitative agreement with experimental nuclear binding energies is not the goal at this exploratory stage; the framework reproduces the proton–electron model’s qualitative predictions through the same RealQM machinery developed for chemistry.

**Extension to heavier nuclei.** The He-4 setup — 2 electrons inside, 4 protons outside, with the 2:1 ratio fixed by the mutual-confinement requirement — generalises to any stable nucleus with  $N \approx Z$  by scaling both species in proportion. The corresponding configurations for the next few  $N=Z$  nuclei are tabulated below, each with electrons placed at the vertices of a regular polytope inside and protons at a larger polytope outside, locking the inner–outer Bernoulli boundary as in the He-4 case.

| Nucleus | Z | N | Electrons (inner geometry) | Protons (outer geometry)                |
|---------|---|---|----------------------------|-----------------------------------------|
| He-4    | 2 | 2 | 2 (linear pair)            | 4 (tetrahedron)                         |
| Li-6    | 3 | 3 | 3 (equilateral triangle)   | 6 (octahedron)                          |
| Be-8    | 4 | 4 | 4 (tetrahedron)            | 8 (cube)                                |
| C-12    | 6 | 6 | 6 (octahedron)             | 12 (icosahedron)                        |
| O-16    | 8 | 8 | 8 (cube)                   | 16 (snub cube or two-shell arrangement) |

For each entry the inner-electron polytope and outer-proton polytope are chosen as nested polyhedra with matching point-group symmetry, which keeps the locked inner–outer Bernoulli boundary structurally similar across the series. The setup runs in the same molecule.js implementation that handles the chemistry validations, with only the particle counts and initial geometry differing.

**Convergence and the early-stopping rule.** The multiphase Coulomb model has a real binding minimum but no built-in floor to prevent post-minimum drift into singular densities. Imaginary-time propagation accordingly finds the variational binding state and then continues toward unphysical collapse if iterated indefinitely. The remedy is an early-stopping criterion: terminate ITP when the per-step energy change falls below a coarse threshold ( $\sim 0.05$  Ha for three consecutive frames) that captures the meaningful binding state before drift sets in. With this rule the configurations above converge cleanly to stable binding energies:

| Nucleus | A  | $E$ (Ha) | $E/A$ (Ha) | $E/A$ rescaled (MeV) | Exp. $E/A$ (MeV) |
|---------|----|----------|------------|----------------------|------------------|
| Li-6    | 6  | −9.57    | −1.60      | −4.34                | 5.33             |
| Be-8    | 8  | −25.88   | −3.24      | −8.80                | 7.06             |
| C-12    | 12 | −25.90   | −2.16      | −5.87                | 7.68             |

**Scale rescaling: from atomic to nuclear units.** The simulations above run at the atomic-physics scale used throughout this paper: lengths in Bohr radii ( $a_0 \approx 0.53 \text{ \AA} = 5.3 \times 10^{-11} \text{ m}$ ), energies in Hartree ( $1 \text{ Ha} = 27.2 \text{ eV}$ ). Atomic nuclei sit at femtometer scales ( $1 \text{ fm} = 10^{-15} \text{ m}$ ), so the length-scale ratio is  $a_0/\text{fm} \approx 5 \times 10^4$ , of order  $10^5$ . Coulomb energy scales as  $1/r$ , so a  $10^5$  length-scale reduction multiplies energies by  $10^5$ . The geometric configurations of the proton-electron model are scale-invariant under this transformation: the same multiphase decomposition, the same locked-boundary topology, the same variational minimum at the same dimensionless inner/outer ratio. Only the dimensional units change. Multiplying our converged  $E/A$  values by  $10^5$  rescales them from Hartree-per-nucleon at atomic scales to electron-Volts-per-nucleon at femtometer scales, then divided by  $10^6$  to express in MeV; net factor  $10^5 \cdot 27.2 \cdot 10^{-6} = 2.72$ .

Applied to the table: Li-6 gives  $-4.34 \text{ MeV/nucleon}$  against the experimental  $-5.33 \text{ MeV/nucleon}$  (81%); Be-8 gives  $-8.80 \text{ MeV}$  against  $-7.06$  (125%); C-12 gives  $-5.87 \text{ MeV}$  against  $-7.68$  (76%). Across three nuclei spanning  $Z=3$  to  $Z=6$ , the proton-electron model with pure Coulomb interactions and no strong-force input is within a factor of  $\sim 1.3$  of experimental nuclear binding energies per nucleon. The Li-6  $\rightarrow$  Be-8 jump is captured correctly (Be-8 binds more strongly per nucleon than Li-6 in both model and experiment); the Be-8  $\rightarrow$  C-12 step is inverted in the model (slight decrease instead of slight increase), which we attribute to electron-electron crowding at the central octahedron in our hand-tuned geometry rather than to a fundamental physical deficit.

**Significance of the scale rescaling.** The factor- $10^5$  rescaling is not an empirical fit. Coulomb’s law is scale-invariant in the sense that  $V(r) = 1/r$  contains no intrinsic length, so the variational problem on a scaled domain produces the scaled energies exactly. The same continuum-mechanics machinery developed for atomic and molecular chemistry thereby predicts MeV-scale nuclear binding energies per nucleon when the geometric setup is interpreted at femtometer scales, with quantitative agreement within  $\sim 25\%$  across Li-6, Be-8, and C-12.

**Equal mass is what makes the geometry work — and why there is no atomic analog.** The model assigns equal unit mass  $m_e = m_p = 1$  to electrons and protons, so the kinetic-energy operator is identical on both species’ densities. This is not a convenience: it is what allows the inner electron core and the outer proton shell to occupy spatial domains of *comparable extent*, separated by the Bernoulli interface that the mutual-confinement argument requires. With comparable kinetic-energy operators, neither species can collapse to a point relative to the other, and the closed positive cage / closed negative core geometry becomes a stable variational extremum.

A natural follow-up question is whether the model has a direct atomic-scale analog — after all, the molecular cation  $\text{H}_3^+$  (3 protons + 2 electrons) shares the constituent count of the RealNucleus Li-6 setup, and pure Coulomb  $1/r$  is scale-invariant. The answer is no, and the obstruction is deeper than just the mass ratio. *Atomic physics has no comparable object at all*: the kernel of an atom is treated as a structureless point charge  $+Z$ , with no internal geometry. Every quantity computed in standard atomic and molecular quantum mechanics — orbital energies, bond strengths, dipole moments — factors out whatever is happening inside the kernel and replaces it with a single number. The RealNucleus model is precisely a model of *what the kernel actually is*: a multiphase Coulomb configuration of protons and electrons with internal Bernoulli boundaries. There is no atomic-physics counterpart to that internal structure, because atomic physics never asks the question. The proton-electron picture lives at the nuclear scale because that is the only scale at which the internal structure of a positively charged “kernel” is the object of study.

The mass ratio reinforces this point. In real atomic physics  $m_p/m_e \approx 1836$ , so the proton kinetic-energy term is suppressed by that factor, the proton density localises to roughly  $10^{-3}$  of the electron domain, and any internal-structure question collapses to the point-charge picture. The equal-mass choice  $m_e = m_p = 1$  that the RealNucleus model adopts is what lets electron



and proton densities occupy domains of comparable extent and produces the mutual-confinement geometry. That choice is appropriate to the nuclear regime — where strong binding makes the kinetic-energy difference between bare proton and bare electron irrelevant compared to the Coulomb scale — and would be physically wrong at atomic scales. The geometric structure of the proton-electron nucleus is therefore scale-specific in both directions: there is no atomic-scale object it corresponds to, and atomic physics has no machinery that would describe such an object even if it existed.

**An aside on the mass ratio in Schrödinger’s equation.** The factor  $m_p/m_e \approx 1836$  that suppresses the proton kinetic term in the standard Schrödinger equation has no *a priori* physical justification. Coulomb forces  $V(r) = q_1 q_2 / r$  depend only on charges and distances, not on masses. The choice to give the proton its empirical inertial mass in a quantum-mechanical equation governed entirely by Coulomb interactions is an empirical input, not a derivation from a deeper principle. Different mass assignments produce different geometries: the empirical 1836 : 1 ratio collapses the proton density to a point at atomic scales and recovers the standard atomic picture, while equal masses produce the comparable-domain geometry at nuclear scales. Both are mathematically consistent with Schrödinger’s equation; neither is forced by the Coulomb force law. The RealNucleus model uses the equal-mass choice because it is the regime in which the proton-electron pair forms an extended structure of nuclear-scale geometry; the standard atomic equation uses the 1836 : 1 ratio because it is the regime in which the proton can be treated as a stationary point charge. The conceptual point is that the mass ratio in the kinetic-energy operator is a *scale-selection parameter*, not a fundamental input. It selects which geometric regime — atomic or nuclear — the equation is being applied to.

**Direct-simulation binding ladder: saturation from one scale.** Before the reduced single-shell formula, the light nuclei are computed *directly* as full multiphase configurations, built by adding one (2e + 4p) alpha-unit at a time — electron pairs inside, proton quartets outside. With a single energy scale fixed on the deuteron ( $E(^2\text{H}) = -2.12$  model units  $\leftrightarrow$  2.22 MeV) and nothing else fitted, the binding *per nucleon* follows the empirical curve:

| nucleus         | pattern (e   p)   | $E$ (model) | MeV/ $A$ | % exp. |
|-----------------|-------------------|-------------|----------|--------|
| $^2\text{H}$    | 1   2             | −2.12       | 1.1      | calib. |
| $^4\text{He}$   | 2   4             | −19.4       | 5.1      | 72%    |
| $^8\text{Be}$   | 2+2   4+4         | −57.6       | 7.5      | 107%   |
| $^{12}\text{C}$ | 2+2+2   4+4+4     | −94.3       | 8.2      | 107%   |
| $^{16}\text{O}$ | 2+2+2+2   4+4+4+4 | −132.1      | 8.6      | 108%   |

(experimental MeV/ $A$ :  $^4\text{He}$  7.07,  $^8\text{Be}$  7.06,  $^{12}\text{C}$  7.68,  $^{16}\text{O}$  7.98.) The table exhibits the empirical hallmark of nuclear binding — **saturation**: from  $^8\text{Be}$  onward the model holds a uniform 107–108% of experiment at nearly constant MeV/ $A$ , and even tracks the experimental *rise* toward the iron peak ( $7.5 \rightarrow 8.2 \rightarrow 8.6$  vs  $7.1 \rightarrow 7.7 \rightarrow 8.0$ ). Each added alpha-unit contributes a near-constant binding, so saturation emerges from dual Coulomb confinement with one scale and no fitting — without the short-range-force saturation postulate the standard picture requires. Two members sit apart, both correctly: the single-unit  $^4\text{He}$  wants a tighter spacing (returning 103% of its 28.3 MeV experimental binding at its own scale), and the deuteron anchor is anomalously weakly bound, exactly as the real deuteron is. Direct computation beyond  $^{16}\text{O}$ , and the magic numbers 20 and above, are left to future work; the heavier nuclei and the full  $B/A$  curve to U-238 are reached by the reduced single-shell formula that follows.

**The universal single-shell packing model.** Beyond the Platonic-vertex range a clean spherically-symmetric reduction captures the entire periodic table from H-2 to U-238 in a single

closed-form equation. Place all  $2Z$  protons in one outer shell of thickness  $d = c/Z$  around an inner electron core of effective radius  $R_e^{\text{eff}}(Z) = R_e + \gamma Z$ , with the proton-shell midradius

$$r^*(Z) = R_e^{\text{eff}}(Z) + d/2 = R_e + \gamma Z + \frac{c}{2Z}. \quad (24)$$

The Gauss-law algebra cancels the leading  $-2Z^2/r^*$  proton-shell attraction against the leading  $+2Z^2/r^*$  intra-shell proton-proton repulsion, leaving the residual  $-Z/r^*$ . Per nucleon (with  $A = 2Z$  for the symmetric case):

$$\boxed{\frac{|E|}{A}(Z) = \frac{1}{2r^*(Z)} = \frac{1}{2R_e + 2\gamma Z + c/Z} \quad (\text{Ha/nucleon}),} \quad (25)$$

converted to MeV by the nuclear-scale factor  $1 \text{ Ha} \rightarrow 2.72 \text{ MeV}$ . The denominator has two  $Z$ -dependent contributions pulling in opposite directions: the term  $c/Z$  shrinks with  $Z$  (favours binding by tightening the proton-shell), while the term  $\gamma Z$  grows with  $Z$  (disfavours binding through electron-core expansion). The two contributions cross at the saddle point

$$Z_{\text{peak}} = \sqrt{\frac{c}{2\gamma}}, \quad (26)$$

which for  $c \approx 0.5$  and  $\gamma \approx 3\text{--}5 \times 10^{-4}$  a.u. gives  $Z_{\text{peak}} \approx 22\text{--}29$ , matching the empirical Fe-Ni binding maximum. The peak is therefore not a piecewise threshold but an emergent saddle of the proton-confinement / core-expansion competition. With three parameters ( $c, R_e, \gamma$ ) the model traces the entire empirical  $B/A$  curve: rising from  $\sim 3$  MeV at very light nuclei through  $\sim 8.8$  MeV at the Fe-Ni peak and declining monotonically toward  $\sim 7.6$  MeV at U-238, in one analytic expression with no piecewise structure.

**Where the two Coulomb repulsions live in the formula.** Both proton-proton and electron-electron repulsion are present in the universal formula, in different ways:

- **Proton-proton intra-shell repulsion is explicit and essential.** The cancellation that gives  $-Z/r^*$  requires summing the  $+P(P-1)/(2r^*)$  intra-shell pp term alongside the  $-PN/r^*$  proton-shell attraction. Without it the model would predict  $-Z^2/r^*$  binding — two orders of magnitude over-bound.
- **Electron-electron repulsion is implicit, encoded in  $R_e^{\text{eff}}(Z)$ .** The constant  $R_e$  is the value the variational minimum picks for the electron-core size when e-e repulsion balances electron kinetic energy and electron-proton attraction. The linear term  $\gamma Z$  is the variational response of the core to growing e-e repulsion as  $N = Z$  increases: keeping the per-nucleon e-e energy  $\sim N^2/(R_e A) \sim N/R_e$  bounded requires  $R_e$  to grow linearly with  $N$ , which is exactly the  $\gamma Z$  contribution. The empirical fit  $\gamma \approx 3\text{--}5 \times 10^{-4}$  a.u. is the slope of this expansion determined by the actual variational minimum.

Both repulsions therefore drive the binding curve — the proton-proton repulsion through the leading cancellation that produces  $-Z/r^*$ , and the electron-electron repulsion through the core-expansion term  $\gamma Z$  that bends the curve over to produce the Fe-Ni peak and the post-peak decline. The reduced formula is the closed-form output of the full multiphase Coulomb variational minimisation, with  $(c, R_e, \gamma)$  as compressed summaries of the kinetic/repulsion balance for protons and electrons. Detailed 3D polyhedral and radial multi-shell simulations include both repulsions explicitly and converge to negative total energies (Li-6, Be-8, C-12 at  $-9.57, -25.88, -25.90$  Ha respectively), confirming variational stability. The reduced formula does not contradict these simulations — it summarises them.

**The fusion–fission shift at Fe-56.** The saddle at  $Z_{\text{peak}} = \sqrt{c/(2\gamma)} \approx 22\text{--}29$  coincides with the empirical Fe-56 / Ni-62 binding maximum and has a direct astrophysical meaning. Below the peak, increasing  $Z$  raises  $|E/A|$ , so combining lighter nuclei into a heavier one releases binding energy — this is the fusion regime, the energy source of stellar nucleosynthesis up to iron. Above the peak, increasing  $Z$  lowers  $|E/A|$ , so a heavy nucleus can release binding energy by splitting into lighter fragments — this is the fission regime, the energy source of heavy-element decay and reactor power. The model identifies the fusion-to-fission shift at Fe-56 as the single point where the proton-confinement term  $c/Z$  stops decreasing  $r^*$  faster than the core-expansion term  $\gamma Z$  increases it. No threshold is imposed; the shift between the two energy-releasing regimes follows from the same closed-form denominator that produces the curve. The astrophysical fact that all elements heavier than iron must be synthesised by neutron-capture processes (not fusion) is therefore a geometric consequence of the  $c/Z$  vs  $\gamma Z$  balance, independent of any strong-force input.

**The geometric essence of the model.** The single-shell reduction makes the model’s structural content transparent: *electrons are large, protons are small*. Each electron’s effective domain inside the core has linear size  $\sim R_e/Z^{1/3}$  (the  $Z$  electrons sharing the core volume), while each proton’s domain has linear size  $\sim c/Z$ . The ratio

$$\frac{\text{electron size}}{\text{proton size}} \sim \frac{R_e Z^{2/3}}{c} \quad (27)$$

grows as  $Z^{2/3}$ , the same scaling that governs surface-to-volume ratios in the Weizsäcker model. Here, however, the  $Z^{2/3}$  dependence arises geometrically from the proton-electron size hierarchy rather than from a postulated surface penalty, and it sets the entire binding curve through the single equation above. The proton-electron model with  $m_e = m_p = 1$  thus produces a quantitative match to the rising plateau of nuclear binding energies from a model containing essentially one free length scale ( $R_e$ ).

**The  $Z^{2/3}$  exponent: two derivations, one number.** The fact that the same  $Z^{2/3}$  (equivalently  $A^{2/3}$ ) exponent appears in both the standard Weizsäcker semi-empirical mass formula and the RealNucleus packing model invites direct comparison, because the two derivations rest on entirely different physical inputs.

In the standard Weizsäcker argument,  $A^{2/3}$  enters as a *surface penalty*. The chain of assumptions is: (i) the strong nuclear force is short-range and saturates, so each nucleon binds a fixed number of neighbors regardless of  $A$ ; (ii) saturation implies constant nuclear matter density and hence  $R \propto A^{1/3}$ ; (iii) the nuclear surface area then scales as  $S \propto A^{2/3}$ ; (iv) surface nucleons have fewer neighbors than interior ones, leading to a binding deficit proportional to  $S$ . Every step in this chain is downstream of the short-range strong-force assumption; remove it and the entire derivation collapses.

The RealNucleus derivation of  $Z^{2/3}$  uses none of these ingredients. There is no short-range force, no saturation, no constant-density assumption, and no  $R \propto A^{1/3}$  ansatz. Instead the exponent arises from the geometric topology of the model: electrons live *inside* as a core of fixed size  $R_e$ , so each of the  $Z$  electrons occupies a volume  $\sim R_e^3/Z$  and has linear size  $\sim R_e/Z^{1/3}$ ; protons live *outside* as a shell of thickness  $d = c/Z$ , so each proton has linear size  $\sim c/Z$ . The ratio of these two scales is  $Z^{2/3}$  by construction. The same exponent then governs the binding-energy plateau, because the proton-shell midradius approaches the constant  $R_e$  at large  $Z$  and the relative correction  $c/(2ZR_e)$  that controls deviations from the plateau also scales with the size-hierarchy parameter.

That two physically incompatible accounts converge on the same exponent is a non-trivial observation. The standard derivation requires short-range strong-force physics to enforce constant density; the RealNucleus derivation requires only pure Coulomb plus the topological choice

of inside-out geometry. Both reproduce the empirical  $A^{2/3}$  scaling. The RealNucleus model therefore exhibits an instance of the Weizsäcker surface term *without* the strong-force premise that the standard textbook derivation treats as essential to it.

**Fusion as a multiphase rearrangement:**  $D + T \rightarrow \text{He-4} + n$ . The proton-electron picture makes one nuclear reaction particularly clean to model:  $D + T \rightarrow \text{He-4} + n + 17.6 \text{ MeV}$ , the canonical fusion used in deuterium-tritium reactors (ITER, NIF). The species involved are: *deuterium* (D, the hydrogen-2 nucleus with 1 proton + 1 neutron); *tritium* (T, the hydrogen-3 nucleus with 1 proton + 2 neutrons); *helium-4* (He-4, the alpha particle, with 2 protons + 2 neutrons); and a free *neutron* (n). Translating each via the proton-electron identification neutron = proton + electron gives the model’s constituent breakdown:

- D = 2 protons + 1 electron (net charge +1)
- T = 3 protons + 2 electrons (net charge +1)
- He-4 = 4 protons + 2 electrons (net charge +2)
- $n = 1$  proton + 1 electron (net charge 0)

Both sides total 5 protons + 3 electrons. The reaction is therefore a *pure topological rearrangement* of the same constituents from two clusters (D + T) into one tight cluster (He-4) plus one weakly bound pair (the neutron) that drifts off. No particle creation, no annihilation, no photon emission.

In a multiphase 3D simulation with all 8 subdomains (5 protons + 3 electrons) initialised in a D-blob + T-blob configuration with all Bernoulli boundaries free and dynamics enabled, the variational principle finds the post-fusion configuration spontaneously: the total energy descends from  $\sim -6$  Ha (separated D + T) to  $\sim -11.5$  Ha (He-4 + ejected neutron) over the imaginary-time relaxation, a drop of  $\Delta E \approx 5.5$  Ha. Rescaled to MeV at the nuclear scale (1 Ha  $\rightarrow$  2.72 MeV), this corresponds to an energy release of  $\Delta E \approx 15$  MeV, compared with the experimental Q-value of 17.6 MeV — agreement to within  $\sim 15\%$  from pure Coulomb with no strong-force input. The smaller static single-shell formula under-predicts this by a further factor of two because it averages over angular structure; the full 3D simulation, with all Bernoulli boundaries free and the angular arrangement of protons around the merged electron core resolved, captures most of the experimental fusion energy. The remaining  $\sim 15\%$  gap is the doubly-magic He-4 closed-shell enhancement not present in the spherically-averaged geometry.

The reaction *itself* occurs without any reaction rule, transition-state ansatz, or strong-force input: the system finds the right answer because the right answer is the variational minimum of 5 protons + 3 electrons in the proton-electron continuum-mechanics framework, namely a tight  $\{4p + 2e\}$  cluster (He-4) plus a free  $\{1p + 1e\}$  pair (the neutron). This demonstrates the model’s reach beyond static binding-energy fits to actual nuclear reactions, using the same multiphase-Bernoulli machinery used elsewhere for chemical reactions (bond breaking and forming).

**Hydrogen burning without neutrinos:**  $4\text{H} \rightarrow \text{He-4}$  as **pure rearrangement**. Stellar hydrogen burning — the net reaction  $4\text{}^1\text{H} \rightarrow \text{}^4\text{He} + 2e^+ + 2\nu + \gamma$  that powers main-sequence stars — is treated in standard nuclear physics as a weak process: two of the four protons must convert to neutrons via  $p \rightarrow n + e^+ + \nu$ , and the emitted neutrinos are the observable signature (the solar-neutrino flux). In the proton-electron picture the reaction is a *pure topological rearrangement* with no weak step. The constituents are  $4\text{H} = 4p + 4e$  and  $\text{}^4\text{He} = 4p + 2e$ ; the reaction  $4\text{H} \rightarrow \text{He-4}$  is simply the regrouping of four proton-electron pairs into a tight  $\{4p + 2e\}$  cluster, releasing two electrons to the surrounding electron sea. Because a neutron is here *already* a bound proton-electron pair, no proton is converted to a neutron and no neutrino is required: the “weak” content of the standard reaction is, in this model, the redistribution of electrons among Coulomb domains. The energy release is the difference between the loosely

bound four-pair configuration and the He-4 minimum, found by the same variational relaxation as the D+T case above. The model thus offers a Coulomb-only reading of the stellar energy source in which the neutrino flux is not a mandatory by-product of fusion but a feature of the specific weak-interaction bookkeeping of the standard picture — a falsifiable structural difference rather than a reinterpretation of the energetics.

**Muon-catalyzed fusion as a mass-scaled domain compression.** The mass-as-scale-selector point (the aside above) makes muon-catalyzed fusion a clean corollary. In the muonic molecular ion  $dd\mu$ , a negative muon ( $m_\mu \approx 207 m_e$ ) replaces the electron binding two deuterons. Since the domain size in the kinetic-energy term scales inversely with the mass, the muonic “bond length” contracts by the mass ratio: the deuteron-deuteron equilibrium separation of the ordinary  $D_2^+$  ion,  $R_e \approx 2.0 a_0$ , shrinks by  $\sim 207$  to a few hundred femtometres, a  $\sim 10^2$ -fold linear compression that brings the two deuterons into fusion range. In RealQM this requires no new physics: rescaling the per-domain mass in the same multiphase solver that computes the  $D_2^+$  potential curve reproduces the muonic contraction directly, and the deuterons are delivered to contact by the ordinary Coulomb variational minimum rather than by tunnelling through a barrier. The example illustrates, at the boundary between the atomic and nuclear regimes, that the geometry of a Coulomb bound state is set by the mass in the kinetic term — the same scale-selection principle that distinguishes the atomic (1836:1) from the nuclear (equal-mass) regime. (Both this and the hydrogen-burning example are Born–Oppenheimer / fixed-grid reductions; they exhibit the scaling and the rearrangement, not a full dynamical fusion rate.)

**Scope: full periodic table, no strong force required.** Across the three reduced models, the proton-electron picture recovers experimental nuclear binding energies per nucleon over the entire periodic table using pure Coulomb interactions and a small number of geometric parameters. No strong-force postulate, no surface-vs-volume balance, no nearest-neighbor saturation, and no fitted asymmetry term are required. The Weizsäcker-style  $Z^{2/3}$  surface scaling emerges geometrically from the proton-electron size hierarchy rather than from a postulated surface penalty. The post-Fe-56 decline emerges from continuous core expansion — the natural geometric response when one proton shell cannot accommodate more constituents at the original packing. The nuclear demonstration is therefore not restricted to fusion-side or light nuclei: the same Coulomb-with-Bernoulli-boundary formulation that handles atoms, molecules, hydrogen bonding, and protein folding accounts for the full  $B/A$  curve through a single mechanism, with no additional force law introduced anywhere in the chain.

## 9 Chemistry vs Physics

A central question in the philosophy of chemistry is whether **(A)** chemistry can be explained by quantum physics, as **Dirac famously claimed in 1929 at the birth of quantum mechanics**, or **(B)** something more is required, as has been argued by leading chemists for nearly a century. Position (B) is the view that chemistry possesses an irreducible conceptual content — orbitals, bonds, valence, hybridisation, electronegativity — that does not follow from Schrödinger’s equation alone, no matter how many computational resources are thrown at it. Position (A) is the view that chemistry *is* applied quantum physics, and that the apparent autonomy of chemical concepts is an artefact of the practical impossibility of solving the underlying equations.

The hundred-year stalemate has been driven less by disagreement on principle than by the fact that, within the standard  $3N$ -dimensional wavefunction formulation that Dirac himself had in mind in 1929, the equations are indeed unsolvable for any system of chemical interest without approximations whose chemistry-specific character (basis sets, exchange–correlation functionals, fitted force fields) seems to vindicate position (B).

RealQM provides **concrete evidence that Dirac’s position (A) was correct, although not within the 1929 wavefunction-on- $3N$ -space formulation that Dirac himself proposed**. The reformulation as a system of non-overlapping unit electron densities in real 3D space, with energy minimization over Coulomb interactions alone, recovers atoms (Section 4.1), the covalent bond and its He limit (Section 4.3), atomization energies of hydrides across periodic-table groups (Section 4.4), reactive chemistry (Section 4.7), and protein folding (Section 4.8) — using only physics, with no chemistry-specific empirical input beyond the kernel parameters that encode the level-by-level reduction. Chemistry emerges as the variational principle’s response to multiple nuclei being present, with no separate machinery required.

The conclusion we draw is that the standard  $3N$ -dimensional formulation was a wrong turn at the foundational level. The configuration-space wave function generates the chemistry-specific apparatus that has, ever since 1929, made chemistry look more autonomous than Dirac’s position would suggest; in the real-3D-space picture Schrödinger envisaged, that apparatus is not needed and chemistry recovers its character as applied physics. The numerical successes of the standard formulation document that the underlying physics — Coulomb interactions plus Schrödinger’s original real-space equation — is right, not that the configuration-space step was a correct foundational choice. Dirac was right about chemistry-as-applied-physics; the equations were just being written in the wrong space.

**Forward look: scaling.** The multiphase decomposition places each one-electron wavefunction on a non-overlapping subdomain of a single 3D grid. Total computational cost is then linear in the number of mesh cells, and hence — at fixed atomic density and fixed mesh resolution matched to the reduced kernel — linear in the number of atoms. This contrasts with standard wavefunction quantum mechanics, whose intrinsic cost is superlinear in electron count regardless of discretisation, since the wavefunction itself lives on  $\mathbb{R}^{3N}$ . With the reduced-kernel mesh native to the model ( $h \sim 0.3$  a.u.), a  $1000^3$  grid accommodates  $\sim 10^6$  atoms within a single workstation-class GPU.

**Forward look: molecular dynamics.** Three properties of the formulation make it directly useful as a force generator for molecular dynamics, beyond what standard quantum mechanics offers. *First*, reactive forces are native: bond breaking and bond formation are simply rearrangements of the multiphase tiling  $\{\Omega_i\}$ , with the same ITP and Hartree machinery used at equilibrium. There is no methodological switch between equilibrium and transition-state regimes, and no need for special-purpose reactive force fields whose calibration regions are necessarily limited. *Second*, the Hellmann–Feynman force on a nucleus is a single central difference of the converged Hartree-plus-kernel potential at that nucleus — the same operation regardless of system size, with none of the basis-set-derivative pathologies (Pulay corrections, basis-set superposition error, polarisation-function sensitivity) that complicate standard *ab initio* molecular dynamics. *Third*, the linear-mesh-cost scaling above places *ab initio* molecular dynamics of  $10^3$ – $10^4$  atom systems — previously a cluster-scale workload — on a single GPU. The implementation already runs interactive Brownian dynamics with full RealQM forces for 343-water clusters (Section 3) and small proteins (Section 4.8), at picosecond-trajectory resolution on consumer hardware. The natural further use is as the upstream parameter source for classical and machine-learned potentials: a single self-consistent quantum treatment of the entire training set, in place of the patchwork of DFT, MP2, CCSD, and experimental fits that current force fields and ML potentials draw from.

**Forward look: from molecules to cells.** A protein of typical size occupies  $10^4$ – $10^5$  atoms; an entire bacterial cell, with its full proteomic content, occupies  $\sim 10^9$ – $10^{10}$  atoms in  $\sim 10^6$ – $10^7$  protein copies drawn from  $\sim 10^3$ – $10^4$  distinct species. The natural way to bridge this scale gap is to use RealQM as the upstream source of *reduced-model parameters*. For each protein species,



a once-and-done RealQM run yields the converged electron density and the molecular Hessian; from these one extracts the solvent-accessible isosurface, the surface electrostatic potential, the hydrophobicity map, and the first  $\sim 10$ – $20$  functional normal modes. The cell is then a Brownian-dynamics gas of such envelopes diffusing, rotating, and interacting through their pre-computed surface fields, with internal flexibility carried by the modes. The same hardware that ran the protein at the foundational level can run the cell at the population level. Concrete predictions accessible in this regime include spatial concentration distributions, encounter and binding kinetics under realistic crowding, signalling-cascade timings, the formation of membraneless condensates, and mutational effects on cell-level phenotype — all derived from a single self-consistent quantum source rather than a stack of empirical and ab-initio fits that often disagree with each other. Whether this programme delivers in practice is an open question; what is no longer open is whether the foundational level is in principle reachable. We do not pursue these extensions here. The intent of the present paper is to establish the variational principle and to demonstrate its predictive content at atomic and small-molecular scale.

## 10 Dissipation, diffusion, and the parabolic structure of RealQM

Standard quantum mechanics is a unitary, time-symmetric theory. The Schrödinger equation conserves energy and is reversible: time can run forward or backward through the same equation. Yet every physical system that quantum mechanics describes — atoms in molecules, electrons in solids, nuclei in chemistry — exhibits irreversible behaviour: heat flows, reactions proceed in one direction, transient excitations decay. The standard reconciliation is statistical-mechanical: trace out an environment (decoherence), average over an ensemble, or impose a thermal-bath coupling (Langevin), and the macroscopic dissipative behaviour emerges from underlying reversible microdynamics.

RealQM has a different starting point. Its dynamical content combines three structurally parabolic ingredients:

- **Imaginary-time projection** on  $\psi_i$ : a heat equation in pseudo-time, projecting onto the local ground state at fixed nuclear positions.
- **Gradient flow** on the energy functional  $E[\{\psi_i\}, \{R_a\}]$ : an intrinsically energy-decreasing dynamics on the slow variables.
- **Level-set free-boundary regularisation** on  $\Gamma_{ij}$ : a parabolic smoothing of the moving electron-domain interfaces, with diffusivity proportional to the local advection driver.

Each of these is intrinsically irreversible. Energy is a Lyapunov function for the gradient flow; the level-set diffusivity smooths small-scale boundary structure; ITP discards high-eigenvalue components in pseudo-time. Time-reversal symmetry is broken not by an added thermal coupling but by the parabolic structure of the equations themselves.

Three observations follow.

**Reactions are captured without statistical input.** An exothermic reaction in RealQM is a transition between two configurations of the multiphase tiling, driven by the gradient of  $E[\{\psi_i\}]$ . The variational structure carries the system from a higher-energy configuration through the saddle to a lower-energy one without any random force, ensemble average, or thermal reservoir. The released energy  $-\Delta H$  is absorbed by the combination of imaginary-time relaxation of  $\psi$  and, when present, velocity damping on the nuclei; both are deterministic. The qualitative reactive event — bond breaking, bond formation, descent to products — is therefore intrinsic, not added on. This is what the existing reactive demonstrations (proton transfer, salt dissolution,  $S_N2$ ) deliver in browser-class compute time.

**Statistical equilibrium still requires statistical input.** Maintaining a system at fixed temperature with the correct fluctuation spectrum requires a random force satisfying the fluctuation–dissipation relation. RealQM does not provide this on its own: the parabolic regularisation is small in magnitude, and its associated thermal scale is many orders of magnitude smaller than what room-temperature thermodynamics demands. To run at finite  $T$  we add a Langevin thermostat, exactly as in classical molecular dynamics. The thermostat represents the surroundings; it is not a fundamental ingredient of the formalism. Empirically the intrinsic level-set viscosity is too weak to thermalise small clusters on phonon timescales — on those scales the dynamics is effectively Hamiltonian. The dissipation is structurally present but quantitatively sub-dominant.

**The unified picture.** RealQM is therefore positioned between the two classical extremes. Standard QM has no dissipation in its core dynamics and must import it via statistics. Classical statistical mechanics has dissipation built into the macroscopic equations but no microscopic content for chemistry. RealQM has both: a microscopic real-space wave-function dynamics that is intrinsically parabolic at the slow-variable level, plus the same Langevin coupling to a reservoir as classical molecular dynamics when finite-temperature equilibrium is desired.

Diffusion — in the precise sense of a parabolic equation governing the slow evolution of a smooth quantity — is therefore not added *on top of* RealQM, the way it is added on top of standard QM through statistical mechanics. It is part of the formalism. The macroscopic heat equation is the natural large-scale limit of the level-set free-boundary motion; the chemical-reaction rate is governed by gradient-flow descent over the energy landscape; vibrational equilibration in dense phases occurs through the same boundary regularisation that stabilises the level-set tracking. The Second Law, the arrow of time, and the irreversibility of chemistry are consequences of the parabolic structure of the dynamical equations, not consequences of an external statistical postulate.

This does not eliminate the need for statistical mechanics — finite-temperature ensemble sampling, fluctuation spectra, and equilibrium constants all still require the canonical machinery. But the relation between quantum mechanics and thermodynamics is reorganised: in standard QM they are conceptually separate, one reversible and one irreversible, connected only through a limit-and-average procedure laid on top. In RealQM they are aspects of the same dissipative variational principle, with statistics needed only to specify the surroundings rather than to introduce irreversibility itself.

**The foundational case for parabolic structure.** The argument has a deeper dimension that bears stating directly. Real-time Schrödinger evolution is unitary as an *idealised* mathematical object: a continuous, infinitely-precise, energy-conserving map of one state to another. But any realisation of that evolution — on any computational substrate, with any finite numerical precision, on any discretisation of space and time, with any finite timestep — introduces small parabolic components from rounding, truncation, and grid spacing. Numerical viscosity cannot be eliminated, only made small. The discrete heat-equation operator  $u_{n+1}(x) = (1 - 2\lambda)u_n(x) + \lambda(u_n(x+h) + u_n(x-h))$  is, by construction, many-to-one: distinct initial profiles having the same local mean evolve to the same updated value. Information is destroyed at every step, not as noise but as a structural consequence of taking a local average. The same operator, applied to  $\psi_i$  rather than to  $u$ , drives the imaginary-time projection at the heart of RealQM and of every quantum Monte Carlo method.

If “physics” means what can actually be computed and observed with finite resources, then physics is intrinsically dissipative, and a formalism whose dynamical structure is parabolic from the start is more faithful to that condition than one whose foundational equations are unitary and whose irreversibility is recovered only through later statistical layers. RealQM’s parabolic structure is on this view not a regularisation choice but a representation-level honesty about the irreducible role of information destruction in any realisable evolution. Irreversibility, the arrow

of time, and the Second Law need not be derived as emergent consequences of microreversible dynamics; they can be read directly off the parabolic equations that any actual computation of physics must employ.

## 11 Two kinds of statistics in quantum mechanics

Standard quantum mechanics carries two distinct statistical commitments that should be separated. The first is **ontological**: the Born rule, by which  $|\psi|^2$  is intrinsically a probability density and single-measurement outcomes are intrinsically random. This is not a description of incomplete information but a claim about the world itself. The second is **bookkeeping**: thermal and ensemble averages, partition functions, density matrices — the canonical machinery of statistical mechanics layered on top of quantum dynamics to handle many-body and finite-temperature systems.

RealQM rejects the first layer outright. The wave function  $\psi_i$  is a deterministic charge density on a non-overlapping spatial domain  $\Omega_i$ ; there are no probabilities and no measurement-induced collapse in the formalism. The dynamics is given by deterministic PDEs — ITP for ground states, gradient flow for slow variables, level-set evolution for boundaries (Section 10).

The second layer is retained only as bookkeeping. Finite-temperature behaviour can be modelled deterministically by coupling the wave equation to a heat field  $T(x, \tau)$  evolving under

$$\frac{\partial T}{\partial \tau} = D \nabla^2 T + Q(x, \tau), \quad Q = -\partial e_{\text{wave}} / \partial \tau, \quad (28)$$

with  $e_{\text{wave}}$  the local energy density of the wave dynamics. The dissipation released by the parabolic ITP relaxation thus feeds a parabolic heat field as a deterministic source term, with exact conservation  $E_{\text{wave}} + E_{\text{thermal}} = \text{const}$ . Equipartition follows from Hamilton’s equations together with Liouville’s theorem and ergodicity rather than from a probability axiom. Temperature, entropy, and free energy  $F = E - TS$  emerge as collective deterministic quantities; ensemble formulas (canonical partition functions, Gibbs measures) are convenient compressions of the same physics, not its substrate.

The substantive claim is therefore: *quantum mechanics does not require ontological statistics*. The Born rule is a feature of one particular interpretation, not of the physical world. The 100-year-old foundational question — *is nature fundamentally random?* — has, on the RealQM reading, a clean negative answer. What remains as “statistical” in RealQM is purely epistemic: incomplete information about initial conditions, and the practical impossibility of tracking  $10^{23}$  deterministic trajectories. The physics underneath is fully deterministic at every level; the statistical character of quantum predictions, where it appears, is a property of our descriptions, not of nature.

**Computational consequences.** The split has direct computational implications. A statistical formulation carries two costs that a deterministic one avoids. *Dimensionality*: the antisymmetric many-body wavefunction in standard QM lives in  $3N$ -dimensional configuration space, exponential in the number of electrons. RealQM, by contrast, uses a single 3-dimensional grid: the total wave function  $\Psi$  is one 3D field that takes the value  $\psi_i$  in each non-overlapping domain  $\Omega_i$ , with the  $N$  electrons partitioning the same physical space rather than living in a higher-dimensional configuration space. For  $N = 50$  electrons, the exact StdQM state space is 150-dimensional and computationally inaccessible without aggressive approximation; the RealQM state is one 3D grid of fixed size, independent of  $N$ , and fits comfortably on a laptop GPU. *Stochastic noise*: statistical methods — Monte Carlo sampling, path-integral QMC, ensemble averaging — have variance scaling as  $1/\sqrt{M}$  in the number of samples  $M$ , so halving the error of any expectation value quadruples the cost. A deterministic formulation produces a clean signal at fixed cost. The asymmetry is not symmetric: a deterministic computation can be augmented

with Langevin random forces at  $O(N)$  extra cost per timestep when finite-temperature behaviour is wanted, while a statistical formulation cannot extract a noise-free signal without paying the variance penalty in full. RealQM thus pays the price of a structural assumption (electrons live in non-overlapping domains) and gains in return both polynomial scaling and deterministic precision — the reason the simulations reported in this paper run in a browser, where StdQM-equivalent calculations require cluster-class compute.

**The double-statistics awkwardness of standard QM thermodynamics.** The two-layer structure of standard quantum statistical mechanics has a conceptual tension that is rarely made explicit. The expectation value at temperature  $T$  is computed as  $\langle A \rangle = \text{Tr}[\rho A]$  with  $\rho = \exp(-\beta H)/Z$ . This combines two distinct probability distributions of *different ontological status*: (i) the Born-rule probability  $|\langle a|n \rangle|^2$  that a measurement on the eigenstate  $|n \rangle$  yields outcome  $a$  — intrinsic and ontological in the Copenhagen reading; (ii) the Boltzmann weight  $P_n = \exp(-\beta E_n)/Z$  giving the thermal probability that the system *is in* state  $|n \rangle$  — epistemic, reflecting incomplete information about which member of an ensemble one happens to have. Stacking the two yields mathematically consistent predictions, but conceptually it is statistics of statistics: an ontological probability averaged with an epistemic one. RealQM, by rejecting the Born rule, removes the first layer outright; thermal statistics is then a single epistemic layer over deterministic configurations. Same predictions, cleaner ontology, no doubled probability.

**Towards a RealQM-based statistical mechanics.** Standard quantum statistical mechanics builds the partition function  $Z(\beta) = \sum_n \exp(-\beta E_n) = \text{Tr} \exp(-\beta H)$  on the eigenvalues of the  $N$ -electron Hamiltonian  $H$ . For realistic many-body systems this spectrum is computationally inaccessible (exponential in  $N$ ), so practical applications resort to model Hamiltonians, mean-field approximations, perturbation theory, or path-integral / quantum Monte Carlo sampling.

RealQM offers a different starting point. The “spectrum” of a system is the set of its **multiphase configurations**  $\{C\}$ : distinct topologies of non-overlapping electron domains  $\{\Omega_i\}$ , each with a deterministic variational energy  $E_C$  obtained by a single ground-state run on a 3D grid. The partition function then takes the same canonical form

$$Z(\beta) = \sum_C \exp(-\beta E_C), \quad \langle A \rangle(T) = \frac{1}{Z} \sum_C A_C \exp(-\beta E_C), \quad (29)$$

but the sum is over real-space tilings rather than over  $3N$ -dim eigenstates. Each  $E_C$  is a browser-class computation; the canonical machinery (Boltzmann weights, free energy  $F = -k_B T \ln Z$ , equilibrium probabilities) sits on top unchanged.

For nuclear motion the existing implementation already realises this scheme deterministically: classical Newton dynamics with a Langevin thermostat (the `USER_DAMPING` and `langevinKT` parameters of `molecule.js`) samples the Boltzmann distribution by ergodic time-averaging, never computing  $Z$  explicitly. The NaCl crystal-melt simulation of Section 4.13 is exactly this in action: every observable reported there (the Lindemann ratio, the Na–Cl radial distribution function, the melting-temperature window) is a thermal average  $\langle A \rangle(T)$  over the canonical ensemble of ionic configurations, computed by time-averaging a single deterministic-plus-Langevin trajectory rather than enumerating states. This is RealQM-based statistical mechanics already in operation, on a 256-ion system, in a browser. The deterministic heat-equation extension of Section 10 adds an intrinsic finite- $T$  mechanism by coupling the wave dynamics to a parabolic heat field  $T(x, \tau)$ , with exact bookkeeping  $E_{\text{wave}} + E_{\text{thermal}} = \text{const}$ .

Two limitations remain. (a) Enumerating multiphase topologies is not yet automated for general systems; in current practice they are constructed manually (halfspace splits, concentric shells, angular sectors) per problem. (b) For protein folding the conformational state space is exponential in residue count, an irreducible scaling that no formalism makes intrinsically easier; the practical route is sampling, e.g. MD trajectories with a thermostat. But the *spectrum* side is

dramatically simplified: each “state” is a 3D field with explicit physical meaning rather than an antisymmetric many-body wave function in  $3N$ -dim configuration space. RealQM thus offers an alternative form of quantum statistical mechanics in which the canonical ensemble retains its standard structure while the underlying spectrum is real-space, deterministic, and tractable.

## 12 Interaction matter–radiation

The RealQM model can be extended to describe interaction matter–radiation following a principle developed in [9]. Both sides of the interaction admit a wave-equation form. On the *matter* side, the time-dependent RealQM equation for each electron  $\psi_j$  reads

$$i\dot{\psi}_j + H_j \psi_j = 0, \quad (30)$$

with the dot denoting differentiation in time and  $H_j$  the single-particle Hamiltonian seen by electron  $j$  (kinetic plus kernel plus the Hartree potential built from the other electrons). This is structurally a wave equation in the same sense that the standard time-dependent Schrödinger equation is a wave equation, an analog of the wave equation  $u_{tt} - u_{xx} = 0$  but first order in time and complex valued. On the *radiation* side, the corresponding wave equation for the field  $u(x, t)$  is the same dissipative classical wave equation that governed Section 10, here with two dissipative channels:

$$u_{tt} - u_{xx} - \gamma u_{ttt} - \delta^2 u_{xxt} = f. \quad (31)$$

The matter-side equation (30) can be expanded with the same two dissipative channels to model interaction matter–radiation symmetrically:

$$i\dot{\psi}_j + H_j \psi_j - \gamma \ddot{\psi}_j - \delta^2 \nabla^2 \dot{\psi}_j = f_j, \quad (32)$$

where  $-\gamma \ddot{\psi}_j$  is the matter-side analog of the Abraham–Lorentz radiation-reaction term (energy emitted as outgoing radiation by an accelerating electronic density),  $-\delta^2 \nabla^2 \dot{\psi}_j$  is the matter-side viscous channel (energy stored as incoherent internal heat at high spatial frequencies), and  $f_j$  is the field forcing on electron  $j$  from incoming radiation. Matter and radiation thus share the same equation structure — a unitary wave equation augmented by identical radiation and viscous dissipation terms — and exchange energy through the matched forcing/reaction channels:  $f$  in (31) is built from the matter dipoles, and  $f_j$  in (32) is built from the radiation field. The system is closed, deterministic, and symmetric in matter and radiation. Here  $u(x, t)$  is the displacement of a vibrating medium representing matter (a forced charged string),  $f$  is incoming radiation forcing,  $-\gamma u_{ttt}$  is the radiation-reaction term on accelerating charge density (Abraham–Lorentz), and  $-\delta^2 u_{xxt}$  is an internal viscous term that converts coherent wave energy into incoherent heat at small spatial scales. The radiation term carries energy out of the medium as outgoing waves; the viscous term *stores* it as heat by damping high-wavenumber modes. The two channels together produce both the Rayleigh–Jeans low-frequency emission and the high-frequency cutoff that ends the ultraviolet catastrophe.

**Energy balance and the second law of radiation.** Multiplying (31) by  $\dot{u}$  and integrating gives an energy balance  $\dot{E} = A - R$  with absorbed energy  $A = \int f \dot{u} dx$  and radiated energy  $R = \gamma \int \ddot{u}^2 dx \geq 0$ , the latter dissipative by the form of the damping term. For incoming radiation  $f$  to actually *heat* the medium ( $\dot{E} > 0$ ) one needs  $A > R$ , i.e. the incoming intensity must exceed the outgoing intensity. This is the second law of radiation: heat flows from hotter to colder, derived directly from the PDE without invoking entropy or photons.

**Rayleigh–Jeans from spectral analysis.** Decomposing  $u$  in Fourier modes  $u_\nu(x, t)$ , each mode is a damped linear oscillator  $\ddot{u}_\nu + \nu^2 u_\nu - \gamma u_\nu^{(3)} = f_\nu$ . Near-resonance analysis at frequency  $\nu$  shows that the radiated intensity satisfies

$$R(\nu, T) \propto \nu^2 T_\nu, \quad (33)$$

which is the Rayleigh–Jeans law, with  $T_\nu \approx \dot{u}_\nu^2$  identified as the temperature of the corresponding mode. This recovers the classical thermal-radiation law as a property of the deterministic PDE, with no probabilistic postulate at any step.

**Ultraviolet cutoff and Wien’s displacement law.** The Rayleigh–Jeans law alone gives an ultraviolet catastrophe ( $\int \nu^2 d\nu \rightarrow \infty$ ). In Planck’s resolution this is removed by quantising radiation as photons of energy  $h\nu$ . In the deterministic picture the cutoff is supplied instead by the viscous damping term  $-\delta^2 u_{xxt}$  in (31), with the temperature-dependent choice

$$\delta = h/T, \quad (34)$$

where  $h$  is the resolution scale of the medium (the smallest coordination length, not Planck’s constant) and  $T$  is the local temperature. A wave at spatial frequency  $\nu$  then sees a damping rate  $\delta^2 \nu^2 = h^2 \nu^2 / T^2$ . Coherent oscillation persists up to a cutoff

$$\nu_{\text{cut}} \approx T/h, \quad (35)$$

above which  $\delta^2 u_{xxt}$  dominates and converts coherent wave energy into incoherent heat. **The cutoff scales linearly with  $T$  — this is Wien’s displacement law derived from the PDE rather than from quantisation.** The Planck spectrum

$$R_{\text{Planck}}(\nu, T) \propto \nu^2 T \cdot \chi(\nu/\nu_{\text{cut}}), \quad \chi(x) \rightarrow 1 \text{ for } x \ll 1, \quad \chi(x) \rightarrow 0 \text{ for } x \gg 1, \quad (36)$$

then emerges as the Rayleigh–Jeans intensity below cutoff times a cutoff factor whose physical origin is the second damping term. The blackbody acts as a high-pass filter that re-emits frequencies below cut-off and stores frequencies above cut-off as heat. Closing the loop, an additional balance equation for  $T$ ,  $\dot{T} = \int (f^2 - \gamma u_{tt}^2) dx$ , makes the system self-consistent: hotter bodies have larger  $\nu_{\text{cut}}$ , hence emit a wider band, hence radiate more total energy, recovering the Stefan–Boltzmann  $T^4$  scaling on integration.

**Photoelectric and Compton effects without photons.** The same wave equation also explains the historically pivotal evidence for photons. The photoelectric effect requires that ejected electrons have kinetic energy proportional to the incident frequency above a threshold, a fact that a linear wave equation alone cannot reproduce but Einstein’s photon model fits cleanly. The viscous term  $-\delta^2 u_{xxt}$ , with  $\delta = h/T$ , supplies exactly the frequency-dependent threshold behaviour: only modes with  $\nu < \nu_{\text{cut}} = T/h$  are sustained as coherent waves, while higher-frequency input is converted to incoherent kinetic motion of the absorbing system. The frequency threshold of the photoelectric effect is therefore a property of the dissipative wave equation, not evidence for a particle nature of light. The Compton effect admits an analogous treatment as resonant inelastic interaction between coherent and incoherent modes. The two phenomena that established the photon picture in 1905 and 1923 thus admit deterministic-PDE explanations consistent with the rest of this paper’s anti-statistical program.

**Three-term energy balance:  $R + H = F$ .** At stationary state the energy balance for the system (31) takes the form

$$R + H = F, \quad R = \int \gamma u_{tt}^2 dxdt, \quad H = \int \delta^2 u_{xt}^2 dxdt, \quad F = \int f^2 dxdt, \quad (37)$$



expressing that incident energy  $F$  is either re-emitted as outgoing radiation  $R$  or stored as internal heat  $H$ . The two channels switch at the cutoff: below  $\nu_{\text{cut}}$  the radiation reaction dominates,  $R \approx F$  and the body re-emits; above  $\nu_{\text{cut}}$  the viscous term dominates,  $H \approx F$  and the body absorbs as heat. The blackbody is therefore literally a high-pass filter: low-frequency input passes through, high-frequency input is converted to internal heat.

**Universality (Kirchhoff).** Different bodies have different parameters  $(\gamma, \delta)$ , but the standard *reference* blackbody is defined by the universal choice  $\gamma$  maximal (full Abraham–Lorentz radiation reaction) and  $h$  minimal (smallest coordination length / finest precision). All bodies that achieve these limits radiate the same  $R(\nu, T)$  at given  $\nu$  and  $T$  — this is Kirchhoff’s universality of blackbody radiation, recovered here as a property of the limiting choice of PDE parameters rather than as a separate postulate. Real bodies are “greybodies” with smaller  $\gamma$  and/or larger  $h$ ; their emission is bounded above by the universal Planck curve.

**Two-body radiative heat transfer.** For two bodies  $B$  and  $\bar{B}$  in radiative contact (each obeying (31) with the other’s emitted field as forcing), the difference field  $W = u - \bar{u}$  satisfies the homogeneous damped equation  $W_{tt} - W_{xx} - \gamma W_{ttt} = 0$ . Multiplying by  $W_t$  and integrating gives

$$G(t) = G(0) - \int_0^t \int \gamma W_{tt}^2 dx ds, \quad G(t) = \frac{1}{2} \int (W_t^2 + W_x^2) dx, \quad (38)$$

so  $G(t)$  decays monotonically: the temperature difference between the two bodies decreases under radiative contact. Net heat flows from warmer to cooler. The 2nd Law of radiation is thereby derived as a theorem about the deterministic PDE pair, not posited as an independent statistical principle.

**Photoelectric effect without photons.** The same model with a frequency-dependent dissipation term reproduces Einstein’s photoelectric law  $K + W = h\nu$  (without postulating photons). Replace  $-\delta^2 u_{xxt}$  by a non-linear viscosity acting only on modes whose ratio of accelerations to velocities exceeds a threshold:

$$\delta^2(u_\nu) = \alpha(T) \left( \frac{h |\ddot{u}_\nu|}{|\dot{u}_\nu|} - W \right)_+, \quad (39)$$

where  $(\cdot)_+$  denotes the positive part. Since  $|\ddot{u}_\nu|/|\dot{u}_\nu| \approx \nu$ , the term activates exactly when  $h\nu > W$  — the photoelectric threshold. The energy dissipated above threshold corresponds to the kinetic energy of ejected electrons, scaling linearly with  $\nu$  above the critical frequency  $\nu_{\text{crit}} = W/h$ , in agreement with Einstein’s 1905 law and Millikan’s 1916 measurements [9]. The frequency threshold of the photoelectric effect is therefore a property of the dissipative wave equation, not evidence for a particle nature of light. The Compton effect admits an analogous resonant-inelastic treatment. The two phenomena that historically established the photon picture in 1905 and 1923 thus admit deterministic-PDE explanations consistent with the rest of this paper’s anti-statistical program.

**Connection to RealQM.** The structural parallel is exact: in both cases the move is to replace a probabilistic statistical postulate (Born rule for matter; photon statistics for radiation) with a deterministic continuum equation plus a regularisation length/precision scale (kernel softening  $r_c$  for matter; numerical resolution  $h$  for radiation), and to recover the macroscopic law as a property of the regularised PDE rather than as the average of an underlying random process. RealQM and the blackbody wave model are siblings of the same anti-statistical program. A natural future direction is the unified system that couples electrons under the RealQM PDE to the radiation field (31): matter and radiation, deterministic throughout, with all thermal and spectral phenomena emerging from the equations themselves.

## 13 The collaboration as part of the contribution

The mathematical reformulation in Section 2 and the hierarchy of Levels 1–4 are the work of the human author. They predate the AI collaboration and have been developed over a decade of independent work. The author also wrote the initial p5.js prototype simulations that established the numerical core (real-space grid, ITP, Poisson solver, basic visualization) used as the starting point for the WebGPU implementation. The full WebGPU port, the validation suite, the systematic kernel-architecture sweeps reported in Section 4, and the curated public Gallery were developed from those starting templates through extended collaboration with **Claude**, an AI code assistant produced by Anthropic.

We document the pattern of the collaboration here because we believe it is a worth-reporting example of how a single mathematical mind can produce research-grade interactive scientific software with an AI engineering partner.

The human posed mathematical and chemical questions: “What if the model uses  $Z = 2$  instead of  $Z = 4$  for carbon?” “Does the H-bond force correctly point toward the acceptor lone pair?” “Why does this geometry prefer stretched over bonded?” The AI translated these into runnable simulations, identified bugs in real time, ran systematic parameter sweeps, computed and tabulated results, drafted candidate gallery cards, and challenged claims that did not survive scrutiny.

The interaction was iterative and surprisingly productive: a question asked at 9:00 might be answered with a working scan file at 9:05, partial results at 9:30, a full sweep table at 10:00, and a refined analysis incorporating the latest data by 10:30. Across many such cycles, the validation table reported in Section 4 was assembled.

We also record honestly where the AI’s interpretations needed correction. Several times the AI prematurely concluded that a result was “within  $X\%$  of experiment” before convergence had been confirmed; in each case the human pushed back and the AI revised. The Gallery’s “Assessment by Claude” card reflects this — its caveats were rewritten more than once during the project as the evidence accumulated.

We find that the right framing for the collaboration is **mathematical mind + AI engineering partner**: the human supplies the theory, the questions, the chemical intuition, and the standards for what counts as evidence; the AI supplies the implementation, the systematic exploration, and the speed. Neither could have produced this result alone in any reasonable timeframe.

## 14 Code and reproducibility

The full RealQM implementation, including the validation suite, the systematic kernel-architecture sweeps reported in Section 4, and the curated public Gallery, is available at:

- **Code repository:** <https://github.com/Claes542/RealMolecule>
- **Interactive Gallery:** <https://claes542.github.io/RealMolecule/gallery.html>
- **RealQM project site:** <https://physicalquantummechanics.blogspot.com>
- **Author’s blog:** <https://claesjohnson.blogspot.com>
- **Browser requirements:** Chrome 113+, Edge 113+, or Safari 17+ (WebGPU)
- **Hardware:** any modern integrated or discrete GPU ( $\sim 1$  GB GPU memory for  $\sim 200^3$  grid)

All results in this paper can be reproduced by opening the relevant `.html` files in the repository. Each binding-energy data point in Section 4 corresponds to a specific URL parameterization

( $R$ ,  $r_c$ ) of a specific scan file (e.g., `mol_fast_H4X_Z2_scan.html?R=1.0&rc=0.2`). The Gallery’s “Kernel Splitting” and “Periodic-Table Coverage” cards link directly to the scan files used to generate the reported numbers.

We deliberately do not include figures in this preprint. The Gallery contains many interactive visualizations — density slices, 3D molecular views, force-arrow displays, real-time dynamics — that are far more informative as live simulations than as static images. Readers wishing to inspect electron densities, watch molecules relax, or run their own kernel-architecture sweeps should consult the Gallery directly.

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## Competing interests

None declared.

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